Theory and Methods in Causal Inference

Lecture Notes

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STAT 5900B

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Contents

Preface

These lecture notes accompany the graduate course **STAT 5900B: Theory and Methods in Causal Inference** at the Department of Statistics, Iowa State University.

The course targets graduate students in statistics. It assumes familiarity with probability theory, mathematical statistics, and linear models at the level of a graduate sequence, but no prior exposure to causal inference.

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Organization

The notes are organized in four parts.

Part I — Foundations (Chapters 1–4) builds the graphical and conceptual foundation: the causal trinity (SEM, DAG, potential outcomes), DAGs and d-separation, the do-calculus and identification criteria, and the connection between potential outcomes and the do-calculus.

Part II — Identification (Chapters 5–8) bridges the graphical framework and observed data: randomization and back-door adjustment, propensity score methods, instrumental variables, and mediation and front-door identification.

Part III — Estimation (Chapters 9–12) covers semiparametric efficiency theory: estimating equations and influence functions, doubly robust estimation, flexible nuisance estimation with cross-fitting, and estimation under instrumental variables.

Part IV — Policy Learning and Sequential Decision Making (Chapters 13–15) covers policy evaluation, optimal policy learning, and dynamic treatment regimes.

Notation

Throughout these notes:

- $\mathrm{do}(T = t)$ denotes the do-operator (intervention).
- $Y(t)$ denotes the potential outcome under $T = t$.
- $\mathbb{E}[\,\cdot\,]$ denotes expectation; $\perp\!\!\!\perp$ denotes conditional independence.
- $\mathcal{G}$ denotes a DAG; $\mathcal{G}_{\overline{T}}$ denotes the mutilated graph after intervening on T.

Chapter 1

Introduction

Learning Objectives

By the end of this chapter, students should be able to:

1. Articulate the distinction between an interventional distribution $f(y \mid \mathrm{do}(T{=}t))$ and an observational conditional distribution $f(y \mid T{=}t)$, and explain why they differ whenever T is endogenous.
2. Describe the same causal question in all three languages — SEM, DAG graph surgery, and potential outcomes — and begin to translate between them.
3. Derive the two densities explicitly in the Gaussian linear confounded model, and identify the endogeneity bias $\alpha\gamma\sigma_U^2/\sigma_T^2$ as the gap between them.
4. State the two-step paradigm (identification then estimation) and locate each step in the causal inference pipeline.

This chapter is an orientation, not a complete treatment. It introduces four ideas — the interventional/observational distinction, the causal trinity, the Gaussian confounded model as a running example, and the identification-versus-estimation paradigm — that will be developed rigorously in Chapters 2–4 and beyond.

1.1 Motivation: Why Causal Inference Is Hard

Causal inference studies when and how the observational distribution can be used to recover the distribution that would arise under interventions.

1.1.1 The Central Question

Causal inference is concerned with a deceptively simple question: what would happen to Y if we were to *set* $T = t$? This is fundamentally different from the question that standard statistical procedures answer: what is the distribution of Y among units *observed* to have $T = t$?

Definition: Interventional vs. Observational Distribution

- The *interventional distribution* $f(y \mid \mathrm{do}(T{=}t))$ is the distribution of Y in a hypothetical world where T has been *externally set* to t, regardless of the usual data-generating mechanism for T.
- The *observational conditional distribution* $f(y \mid T{=}t)$ is the distribution of Y among the subpopulation of units for whom $T = t$ was *observed* to hold.

These two distributions coincide only when T is randomized (exogenous). In all other cases they differ, and the difference is *endogeneity bias*.

Definition: Exogeneity and Endogeneity

Suppose the outcome is generated by the structural equation $Y = f_Y(T, X, U)$, where T is the treatment, X are observed covariates, and U represents unobserved common causes of T and Y (confounders).

- The treatment T is *exogenous given* X if $T \perp\!\!\!\perp U \mid X$.
- The treatment T is *endogenous given* X if $T \not\perp\!\!\!\perp U \mid X$, so that some unobserved variable U affects both T and Y.

Example: Drug Trial with Unmeasured Severity

Does a new drug ($T = 1$) reduce blood pressure (Y)? Suppose sicker patients are more likely to take the drug, so that unmeasured severity U is a common cause of T and Y. Then:

- *Observational study*: $\mathbb{E}[Y \mid T{=}1] > \mathbb{E}[Y \mid T{=}0]$. Drug users have higher blood pressure on average. A naïve analyst concludes the drug is harmful.
- *Randomized trial*: $\mathbb{E}[Y \mid \mathrm{do}(T{=}1)] < \mathbb{E}[Y \mid \mathrm{do}(T{=}0)]$. Forcing everyone to take the drug reduces blood pressure. The drug is beneficial.

Same data. Different questions. Opposite answers. The entire discrepancy is caused by the back-door path $T \leftarrow U \rightarrow Y$, which the observational quantity $\mathbb{E}[Y \mid T{=}1]$ cannot remove.

The Fundamental Problem of Causal Inference

We observe $f(y \mid T{=}t)$ directly from data. We want $f(y \mid \mathrm{do}(T{=}t))$. The goal of *identification theory* is to find conditions under which the former can be used to recover the latter.

1.2 The Causal Trinity: Three Languages for One Idea

The same causal question — what happens when we set $T = t$? — can be expressed in three distinct but equivalent languages. Collectively, we call these the **causal trinity**: the *structural equation model* (SEM), the *directed acyclic graph* (DAG) with graph surgery, and the *potential outcomes* framework. Figure 1.1 depicts their relationships.

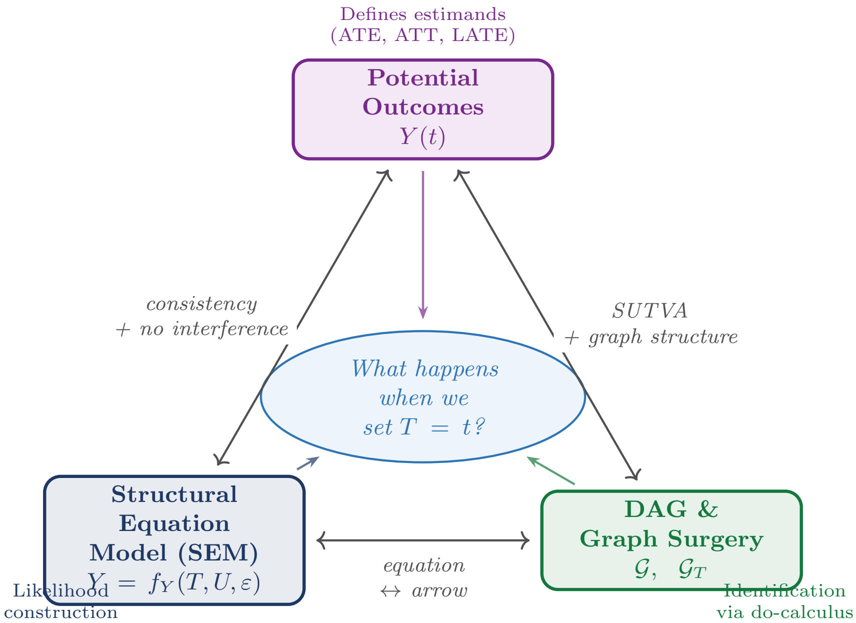


Figure 1.1: The **causal trinity**: three languages that express the same causal question from different vantage points. Potential outcomes define *what* we want to estimate; the DAG with graph surgery specifies *how* to identify it; the SEM provides the *generative mechanism* for likelihood construction.

1.2.1 Language 1: Structural Equation Models

Definition: Structural Equation Model (SEM)

A recursive SEM over variables (U, T, Y) is a system

$$U \sim p(u), \qquad T = f_T(U, \delta), \qquad Y = f_Y(T, U, \varepsilon),$$

where δ and ε are mutually independent exogenous disturbances and U is a confounding factor.

Remark: Error Independence in the NPSEM-IE

The assumption that δ and ε are mutually independent is a substantive restriction. Because U is unobserved and enters both f_T and f_Y, error independence does not remove the dependence between T and $Y(t)$ induced by U; it restricts only the additional channel of dependence running through the equation-specific errors. Richardson and Robins (2014) call SEMs with this structure *Nonparametric Structural Equation Models with Independent Errors* (NPSEM-IEs).
These notes adopt the NPSEM-IE framework throughout: error independence ensures the DAG's Markov condition holds, making the Markov factorization of the joint density immediate.

Observational regime. T is determined by f_T; because U enters both f_T and f_Y, we have $\mathrm{Cov}(T, \varepsilon) \neq 0$ in general: this is endogeneity.

Interventional regime. The do-operator replaces the equation for T with a constant t, yielding the *mutilated system*:

$$U \sim p(u), \qquad T = t \text{ (fixed)}, \qquad Y = f_Y(t, U, \varepsilon).$$

We define the *potential outcome* as $Y(t) := f_Y(t, U, \varepsilon)$; by construction, $Y(t) \sim f(y \mid \mathrm{do}(T{=}t))$.

1.2.2 Language 2: Directed Acyclic Graphs

Definition: Graph Surgery

In the observational DAG, arrows into T encode the mechanisms that determine T. Under $\mathrm{do}(T{=}t)$, all arrows into T are *deleted*. The resulting mutilated graph $\mathcal{G}_{\overline{T}}$ has no back-door paths from T to Y: the causal effect flows only along the direct path $T \to Y$.

1.2.3 Language 3: Potential Outcomes

Definition: Potential Outcome

(Rubin 1974) In the SEM framework, $Y(t) = f_Y(t, U, \varepsilon)$ is the solution for Y in the mutilated system; it is the outcome that *would have been observed* had T been set to t, possibly contrary to fact.
SUTVA (stable unit treatment value assumption): $Y_i = Y_i(T_i)$, i.e., the observed outcome for unit i equals the potential outcome at the actual treatment received. This rules out interference between units and multiple versions of treatment.

1.2.4 Roadmap: How the Trinity Organizes These Notes

Part I — Foundations: the language of DAGs. DAGs and the do-operator are the primary language here because they make the interventional/observational distinction *syntactically explicit*: confounding is visible as an open back-door path, and identifying assumptions are visible as graph criteria.

Part II — Identification: bridging DAGs and observed data. Ignorability $(Y(t) \perp\!\!\!\perp T \mid X)$ is the potential-outcomes counterpart of the back-door criterion; the propensity score $\pi(X) = P(T{=}1 \mid X)$ operationalizes it in observed data.

Part III — Estimation: semiparametric and machine-learning methods. The semiparametric efficiency framework — efficient influence functions, doubly robust estimators, double machine learning — is largely language-neutral.

1.2.5 Historical Roots of the Causal Trinity

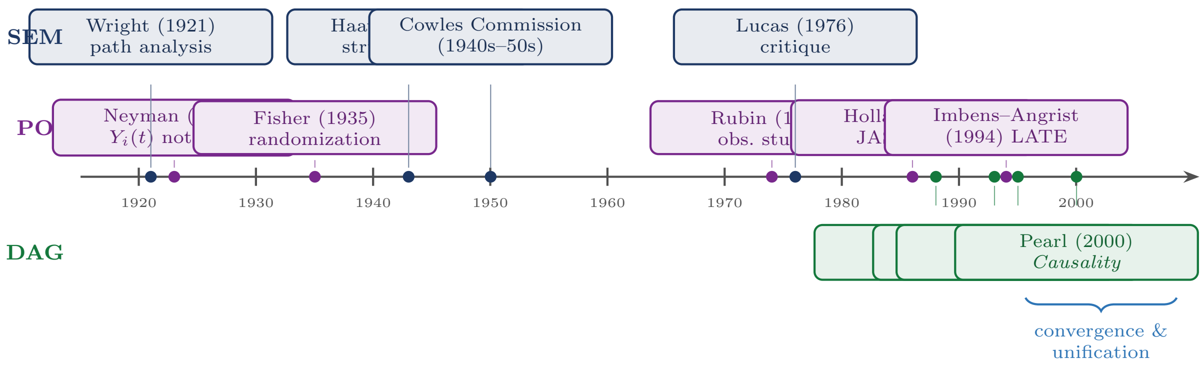


Figure 1.2: Key milestones in the three schools of causal inference. SEM (top, blue) originates in genetics and econometrics; PO (middle, purple) in experimental statistics; DAG (bottom, green) in computer science and philosophy.

1.3 The Core Distinction

We work within the SEM, now augmented with an observed exogenous variable X:

$$T = f_T(X, U, \delta), \qquad Y = f_Y(T, X, U, \varepsilon). \tag{1.1}$$

The single most important inequality of this course is:

$$\underbrace{f(y \mid x,\, \mathrm{do}(T{=}t))}_{\text{interventional}} \neq \underbrace{f(y \mid x,\, T{=}t)}_{\text{observational}} \qquad \text{whenever } T \text{ is endogenous.} \tag{1.2}$$

Definition: Conditional Exchangeability

Given observed covariates X and confounders U, the treatment assignment is *conditionally exchangeable with the intervention* if $Y(t) \perp\!\!\!\perp T \mid X, U$, or equivalently,

$$f(y \mid x,\, T{=}t,\, U{=}u) \;=\; f(y \mid x,\, \mathrm{do}(T{=}t),\, U{=}u).$$

This says that once we condition on all common causes (X, U) of T and Y, observing $T{=}t$ and intervening to set $T{=}t$ produce the same distribution of Y.

Definition: Positivity

The treatment assignment satisfies *positivity* if, for all t in the support of T and almost all (x, u):

$$p(T{=}t \mid X{=}x,\, U{=}u) \;>\; 0.$$

Proposition: Back-Door Adjustment Formula

Under conditional exchangeability and positivity,

$$f(y \mid x,\, \mathrm{do}(T{=}t)) \;=\; \int f(y \mid x,\, T{=}t,\, U{=}u)\, p(u \mid x)\, du.$$

Proof

By positivity, $f(y \mid x,\, T{=}t,\, U{=}u)$ is well-defined. By the law of total probability:

$$f(y \mid x,\, \mathrm{do}(T{=}t)) = \int f(y \mid x,\, U{=}u,\, \mathrm{do}(T{=}t))\; p(u \mid x)\; du.$$

The weight is $p(u \mid x)$, not $p(u \mid x, T{=}t)$, because under $\mathrm{do}(T{=}t)$ the treatment carries no information about U. By conditional exchangeability, $f(y \mid x,\, U{=}u,\, \mathrm{do}(T{=}t)) = f(y \mid x,\, T{=}t,\, U{=}u)$. □

Remark: Confounding Discrepancy

What OLS estimates is the observational density, not the interventional density:

$$f(y \mid x,\, T{=}t) = \int f(y \mid x,\, T{=}t,\, U{=}u)\; p(u \mid x,\, T{=}t)\; du.$$

The kernel is the same as in Equation 1.3, but averaged over the *selection-distorted* weight $p(u \mid x, T{=}t)$. Because U and T are dependent through the back-door path, $p(u \mid x, T{=}t) \neq p(u \mid x)$, and the gap is non-zero whenever T is endogenous.

1.3.1 Two Scenarios: Observed vs. Unobserved Confounder

	Scenario 1: U observed	**Scenario 2: U unobserved**
Confounding type	No unmeasured confounding	Unmeasured confounding
Back-door formula usable?	Yes	No: U latent
Identification route	Back-door adjustment / standardization	IV, front-door, ...
Key assumption	Unconfoundedness: $U \perp\!\!\!\perp T \mid X$	Exclusion restriction, monotonicity, ...
Chapters in this course	Chapters 2–6	Chapter 7

Scenario 1: U is observed. The back-door adjustment formula Equation 1.3 is usable from data:

$$f(y \mid x,\, \mathrm{do}(T{=}t)) = \int f(y \mid x,\, T{=}t,\, U{=}u)\; p(u \mid x)\; du. \tag{1.3}$$

A special case is *unconfoundedness*: if $U \perp\!\!\!\perp T \mid X$ already holds observationally, Equation 1.3 reduces to the familiar regression $\mathbb{E}[Y \mid X{=}x, T{=}t]$.

Scenario 2: U is unobserved. Equation 1.3 is not usable. Additional structure beyond (Y, T, X) must be imposed: **instrumental variables** (Chapter 7), the **front-door criterion** (Chapter 3), or **sensitivity analysis**.

1.4 The Gaussian Linear Confounded Model

1.4.1 Setup

Example: Gaussian Linear Confounded Model

The structural equations are

$$Y = \beta T + \gamma U + \varepsilon, \qquad T = \alpha U + \delta, \tag{1.4}$$

where U is an unobserved confounder and the three primitive variables are mutually independent: $U \sim \mathcal{N}(0, \sigma_U^2)$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, $\delta \sim \mathcal{N}(0, \sigma_\delta^2)$.
Here β is the causal effect of T on Y, γ is the direct effect of U on Y, and α is the direct effect of U

on T.

The endogeneity of T is immediate: $\mathrm{Cov}(T, \gamma U + \varepsilon) = \gamma\alpha\sigma_U^2 \neq 0$ when $\alpha \neq 0$ and $\gamma \neq 0$. Consequently, OLS regressing Y on T does not estimate β.

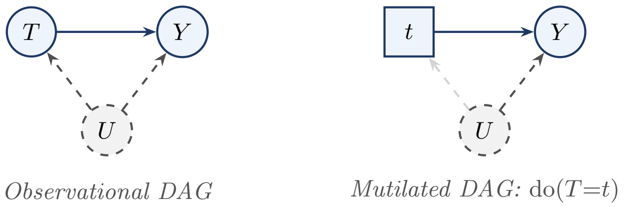


Figure 1.3: Graph surgery on the Gaussian confounded model. The faded arrow in the mutilated DAG has been severed by the intervention $\mathrm{do}(T{=}t)$. The back-door path $T \leftarrow U \rightarrow Y$ is eliminated; only the causal path $T \rightarrow Y$ remains active.

1.4.2 Two Explicit Densities

Write $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$ for the marginal variance of T.

Interventional density. Set $T = t$ externally. U is drawn from its marginal $\mathcal{N}(0, \sigma_U^2)$ independently. Hence

$$f(y \mid \mathrm{do}(T{=}t)) = \mathcal{N}\big(\beta t,\ \ \gamma^2\sigma_U^2 + \sigma_\varepsilon^2\big)\,. \tag{1.5}$$

Observational density. Since (Y, T) is jointly normal with $\mathrm{Cov}(Y, T) = \beta\sigma_T^2 + \gamma\alpha\sigma_U^2$, the conditional normal formula gives:

$$f(y \mid T{=}t) = \mathcal{N}\left(\left(\beta + \frac{\gamma\alpha\sigma_U^2}{\sigma_T^2}\right) t,\ \ \frac{\gamma^2\sigma_U^2\sigma_\delta^2}{\sigma_T^2} + \sigma_\varepsilon^2\right). \tag{1.6}$$

1.4.3 The Endogeneity Gap

	Interventional	**Observational**
Mean	βt	$\left(\beta + \frac{\gamma\alpha\sigma_U^2}{\sigma_T^2}\right) t$
Variance	$\gamma^2\sigma_U^2 + \sigma_\varepsilon^2$	$\frac{\gamma^2\sigma_U^2\sigma_\delta^2}{\sigma_T^2} + \sigma_\varepsilon^2$
Residual	$(\gamma U{+}\varepsilon) \perp\!\!\!\perp T$	$(\gamma U{+}\varepsilon)$ correlated with T
Identified by	RCT (or IV, Chapter 7)	OLS

The *endogeneity bias* of OLS is $\gamma\alpha\sigma_U^2/\sigma_T^2$. Numerically, with $\alpha = \gamma = 1$ and $\sigma_U^2 = \sigma_\delta^2 = 1$: bias $= 1/2 = 0.5$.

1.4.4 Lab: Simulating the Two Densities

The analytical results were verified by simulation using $n = 10{,}000$ draws with parameters $\beta = 2$, $\alpha = 1$, $\gamma = 1$, $\sigma_U = \sigma_\varepsilon = \sigma_\delta = 1$.

Experiment 1: OLS bias.

	Simulated	**Theory**	**Formula**
OLS slope	2.5147	2.5000	$\beta + \gamma\alpha\sigma_U^2/\sigma_T^2$
True β	—	2.0000	β
Bias	0.5147	0.5000	$\gamma\alpha\sigma_U^2/\sigma_T^2$

Experiment 2: Two-sample comparison at $t_0 = 1$.

	Mean (Simulated)	Mean (Theory)	SD (Simulated)	SD (Theory)
$\mathbb{E}[Y \mid T{=}1]$	2.497	2.500	1.229	$\sqrt{1.5} = 1.225$
$\mathbb{E}[Y \mid \mathrm{do}(T{=}1)]$	2.001	2.000	1.428	$\sqrt{2} = 1.414$

Experiment 3: Density overlay for varying α. Figure 1.4 shows the interventional density (solid) and the observational density Equation 1.6 (dashed), both evaluated at $t_0 = 1$, for $\alpha \in \{0, 0.3, 0.7, 1.0\}$.

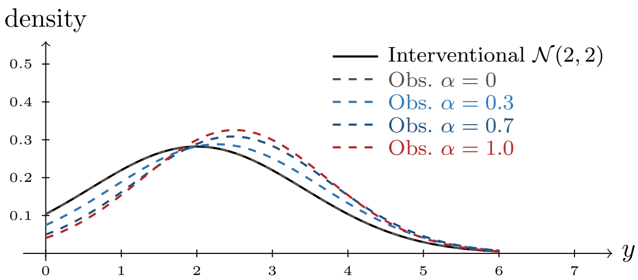


Figure 1.4: Interventional density $\mathcal{N}(2, 2)$ (solid black) versus observational density for $\alpha \in \{0, 0.3, 0.7, 1.0\}$ (dashed), evaluated at $t_0 = 1$. As α increases, the observational curve shifts rightward (growing bias) and narrows (reduced variance).

1.5 The Two-Step Paradigm

Causal inference is a two-step discipline. The two steps are logically distinct and require different tools.

1.5.1 Step 1: Identification

Definition: Identification

A causal quantity $P(y \mid \mathrm{do}(T{=}t))$ is *identified* from the observational distribution P if there exists a functional Ψ such that $P(y \mid \mathrm{do}(T{=}t)) = \Psi(P)$, and $\Psi(P)$ depends only on the observable joint distribution.

Identification is a purely mathematical question about the structure of the causal model: given the assumed graph and assumptions, can the interventional density be expressed as a function of the observed-data distribution? It does not depend on sample size.

The main tools are the back-door criterion (Chapters 2–3, applied in Chapters 5–6), the front-door criterion (Chapter 3), the three rules of the do-calculus (Chapter 3), and IV assumptions (Chapter 7).

1.5.2 Step 2: Estimation

Once $\Psi(P)$ has been identified, the statistical problem is to estimate $\Psi(P)$ from n observations $O_1, \dots, O_n$ as efficiently as possible. The main tools are efficient influence functions (EIF), inverse probability weighting (IPW), doubly robust estimators, and double machine learning (DML), all covered in Part III.

1.5.3 A Worked Example: Flu Vaccination and Infection

Scenario. A public health agency observes 1,000 individuals over one flu season. Each person either received a flu vaccine ($T = 1$) or did not ($T = 0$), and either became infected ($Y = 1$) or did not ($Y = 0$). Age group X (elderly $= E$, young $= Y$) is recorded for all individuals. Elderly people are more likely to be vaccinated and more susceptible to infection, so X confounds the relationship between T and Y.

Observed data.

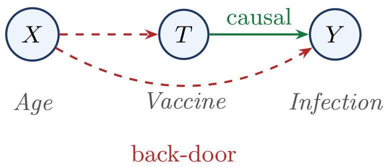


Figure 1.5: Causal DAG for the vaccination example. The green arrow is the causal path $T \to Y$. The red dashed arrows form the back-door path $T \leftarrow X \to Y$.

	Vaccinated $(T = 1)$		**Unvaccinated** $(T = 0)$		Total
	n	Infected	n	Infected	
Elderly $(X = E)$	360	108 (30%)	40	20 (50%)	400
Young $(X = Y)$	60	3 (5%)	540	54 (10%)	600
Total	420	111 (26.4%)	580	74 (12.8%)	1,000

Simpson's paradox.

Stratum	**Vaccinated**	**Unvaccinated**	**Difference**	**Conclusion**
Elderly $(X = E)$	30%	50%	-20 pp	vaccine helps
Young $(X = Y)$	5%	10%	-5 pp	vaccine helps
Aggregate (raw)	**26.4%**	**12.8%**	$+13.6$ pp	**vaccine harms??**

Within every stratum the vaccine reduces infection. The reversal occurs because the vaccinated group is disproportionately elderly (86% vs. 7%).

Step 1: Identification. The set $\{X\}$ satisfies the back-door criterion. By the back-door formula:

$$P(Y{=}1 \mid \mathrm{do}(T{=}t)) = \sum_{x\in\{E,Y\}} P(Y{=}1 \mid T{=}t, X{=}x)\, P(X{=}x). \tag{1.7}$$

Step 2: Estimation.

$$\hat{P}(Y{=}1 \mid \mathrm{do}(T{=}1)) = 0.30 \times 0.4 + 0.05 \times 0.6 = 0.15,$$

$$\hat{P}(Y{=}1 \mid \mathrm{do}(T{=}0)) = 0.50 \times 0.4 + 0.10 \times 0.6 = 0.26.$$

The estimated causal risk difference is $\hat{\Delta}_{\text{causal}} = 0.15 - 0.26 = -0.11$: the vaccine causally reduces infection probability by 11 percentage points.

The Two Steps, Concretely

Step 1 (identification) established — from the DAG alone — that the back-door formula Equation 3.1 expresses the causal quantity as a functional of observable data. **Step 2 (estimation)** replaced the population probabilities with sample proportions.
These two steps are logically separate: Step 1 is a mathematical argument about the causal model that does not depend on sample size; Step 2 is a statistical argument about finite-sample estimation that does not depend on the causal model.

1.6 Summary

1. **Interventional vs. observational distribution.** Causal inference answers questions about interventions $f(y \mid \mathrm{do}(T{=}t))$, not about observations $f(y \mid T{=}t)$. These two distributions coincide only under exogeneity; whenever T is endogenous they differ, and the difference is endogeneity bias.

2. **The causal trinity.** The same causal question can be expressed in three languages — SEM (equation surgery), DAG (graph surgery), and potential outcomes ($Y(t)$). In the SEM framework, $Y(t)$ is defined as the solution for Y in the mutilated system, so $Y(t) \sim f(y \mid \mathrm{do}(T{=}t))$ by construction. Fluency in all three is essential for modern causal reasoning.

3. **Endogeneity bias in the Gaussian confounded model.** In the Gaussian linear confounded model, the coefficient of T in the observational mean $\mathbb{E}[Y \mid T{=}t]$ differs from the causal coefficient β by $\gamma\alpha\sigma_U^2/\sigma_T^2$, where $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$. This is the omitted-variable bias.

4. **Two-step paradigm.** Causal inference is a two-step discipline: *identification* (expressing $\Psi(P)$ as a functional of observable data, a mathematical question about the causal model) followed by *estimation* (constructing $\hat{\Psi}_n$ efficiently from n observations, a statistical question about finite samples). The two steps are logically independent.

Causal inference is the study of when the observational distribution contains enough information to recover the interventional distribution. Everything in this course — the back-door criterion, do-calculus, instrumental variables, doubly robust estimation — is an answer to that question in a particular setting.

1.7 Problems

1. Do-notation fundamentals. For each statement below, decide whether it refers to an interventional or an observational distribution, and rewrite it unambiguously using do-notation where appropriate.

(a) "The probability that a patient recovers given that they took the drug."
(b) "The probability that a patient would recover if we prescribed the drug to everyone."
(c) "Among students who attended tutoring, the average exam score was 85."
(d) "If we enrolled all students in tutoring, the average exam score would be 85."

2. Deriving the observational density. In the Gaussian confounded model, verify Equation 1.6 by carrying out the following steps. Let $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$.

(a) Show that (Y, T) is jointly normal by writing both as linear combinations of the independent normals (U, ε, δ).
(b) Compute $\mathrm{Cov}(Y, T)$ and $\mathrm{Var}(T)$.
(c) Apply the conditional normal formula to derive $\mathbb{E}[Y \mid T{=}t]$ and $\mathrm{Var}[Y \mid T{=}t]$, and hence verify Equation 1.6.
(d) Show that the OLS probability limit is $\beta + \gamma\alpha\sigma_U^2/\sigma_T^2$, and interpret this as an omitted-variable bias formula.

3. The causal trinity. Consider the following verbal causal claim: "Aspirin (T) reduces the risk of heart attack (Y) because it inhibits platelet aggregation (M), but patients with pre-existing cardiovascular disease (U, unobserved) are both more likely to take aspirin and more likely to have a heart attack."

(a) Draw the causal DAG implied by this description.
(b) Write down the recursive SEM with appropriate structural functions f_T, f_M, f_Y.
(c) State the SUTVA assumption and discuss whether it is plausible in this setting.
(d) Explain why $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$ does not identify the causal effect $\mathbb{E}[Y \mid \mathrm{do}(T{=}1)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0)]$ in this DAG.

Chapter 2

DAGs and d-Separation

Learning Objectives

By the end of this chapter, students should be able to:

1. Construct a DAG from a verbal description of a causal model and identify parents, descendants, ancestors, and non-descendants of any node.
2. Classify every intermediate node on a path as a chain, fork, or collider, and apply the blocking rule for each.
3. Apply the d-separation criterion to determine all conditional independence relationships implied by a DAG.
4. Recognize and avoid collider bias, including the case where a descendant of a collider is conditioned upon.
5. State the Markov factorization and use it to write the joint density of a DAG in terms of local conditional densities.
6. Interpret d-separation results as conditional independence assumptions on observed variables, and recognize their role as the graphical expression of causal assumptions.

Readers who want a gentler introduction to conditional independence, the three basic path motifs, and the Markov-property interpretation may consult Appendix A before or alongside this chapter.

2.1 Directed Acyclic Graphs

DAGs allow us to translate qualitative causal assumptions into quantitative statistical restrictions. This single idea underlies everything in this chapter: once we draw a graph encoding what we believe about the causal structure of a problem, the graph tells us exactly which variables are independent, which are confounded, and what we need to condition on to identify a causal effect.

Remark

In this chapter, the word *graph* does not mean a plot of data or a graph of a function. It means a collection of nodes and edges used to represent relationships among variables. Our objective is to understand how such graphs encode statistical structure — especially conditional independence, confounding, and path blocking — and how this graphical structure will later support causal identification.

2.1.1 Basic Definition

Definition: Directed Acyclic Graph

A *directed acyclic graph* (DAG) $\mathcal{G} = (V, E)$ consists of a finite set of nodes V and a set of directed edges $E \subseteq V \times V$ such that there is no directed cycle: no node V_i has a directed path to itself.

In a causal DAG each node represents a variable and each directed edge $A \to B$ encodes "A is a direct cause of B, given all other variables in the model." The acyclicity condition is essential: it guarantees that the variables can be ordered topologically so that the joint density admits the recursive factorization

$$p(v_1, \dots, v_k) = \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i)).$$

2.1.2 Structural Relationships

Definition: Structural Relationships in a DAG

For a node $v \in \mathcal{V}$ in $\mathcal{G} = (\mathcal{V}, E)$:
- $\mathrm{Pa}(v) = \{w \in \mathcal{V} : w \to v \in E\}$ are the *parents* of v.
- $\mathrm{De}(v)$ = all nodes reachable from v by directed paths are the *descendants* of v.
- $\mathrm{An}(v)$ = all nodes with a directed path to v are the *ancestors* of v.
- $\mathrm{Nd}(v) = \mathcal{V}\ \ (\mathrm{De}(v) \cup \{v\})$ are the *non-descendants* of v.

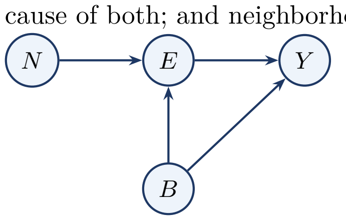
Note that $\mathrm{Nd}(v)$ includes the parents of v; when stating the local Markov property in Section A.4, we will exclude the parents explicitly.

Quick Terminology Summary

A *node* represents a variable, and an *edge* represents a relationship between two variables. If $A \to B$, then A is a *parent* of B and B is a *child* of A. Two nodes are *adjacent* if they are connected by an edge. A *path* is any sequence of connected nodes, regardless of arrow direction; a *directed path* is a path whose arrows all point in the same direction. A *directed acyclic graph* (DAG) is a directed graph with no directed cycles.

Example: Education, Earnings, and Family Background

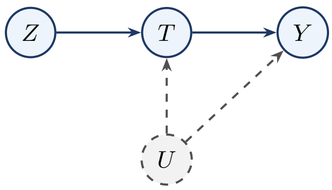
Consider the causal story: education (E) affects earnings (Y); family background (B) is a common cause of both; and neighborhood (N) affects education but has no direct effect on earnings.



The Markov factorization is $p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b)$. Reading off structural relationships: $\mathrm{Pa}(E) = \{N, B\}$, $\mathrm{Pa}(Y) = \{E, B\}$, $\mathrm{Pa}(N) = \mathrm{Pa}(B) = \emptyset$. The descendants of N are $\{E, Y\}$; B and N are non-descendants of each other.

Example: The IV DAG

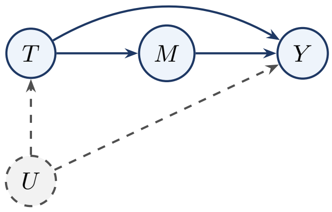
The canonical instrumental variable model involves four nodes: Z (instrument), T (treatment), Y (outcome), and U (unobserved confounder).



$\text{Pa}(T) = \{Z, U\}$, $\text{Pa}(Y) = \{T, U\}$, $\text{Pa}(Z) = \text{Pa}(U) = \emptyset$. The descendants of Z are $\{T, Y\}$; U and Z are non-descendants of each other.

Example: The Mediation DAG

In a mediation model, treatment T affects outcome Y both directly and through an intermediate variable M (the mediator). An unobserved confounder U creates a back-door path $T \leftarrow U \rightarrow Y$, making T endogenous. The mediator M is itself unconfounded.



$\text{Pa}(T) = \{U\}$, $\text{Pa}(M) = \{T\}$, $\text{Pa}(Y) = \{M, T, U\}$, $\text{Pa}(U) = \emptyset$. The total effect of T on Y travels along two paths: the direct path $T \rightarrow Y$ and the indirect path $T \rightarrow M \rightarrow Y$. Because there is a direct path $T \rightarrow Y$ bypassing M, the front-door criterion does not apply here.

2.2 Paths, Blocking, and d-Separation

A DAG encodes direct causal relationships through its edges, but variables can also be statistically related through longer routes that mix directed and reversed edges. The concept of a *path* formalises these routes, and the question of whether a path transmits or blocks statistical dependence — depending on which variables are conditioned upon — is precisely what d-separation answers. The key insight is that conditioning can both *close* channels that were open (by blocking a chain or fork) and *open* channels that were closed (by activating a collider). Figure 2.1 illustrates all three cases.

2.2.1 Path Structures

Definition: Path

A *path* between nodes X and Y in $\mathcal{G}$ is any sequence of distinct nodes $X = V_0, V_1, \dots, V_k = Y$ such that each consecutive pair is connected by an edge in either direction.

Every intermediate node V_m on a path plays one of three structural roles. In a **chain** ($V_{m-1} \rightarrow V_m \rightarrow V_{m+1}$), V_m is a causal intermediary through which influence is transmitted. In a **fork** ($V_{m-1} \leftarrow V_m \rightarrow V_{m+1}$), V_m is a common cause that creates confounding between the two endpoints. In a **collider** ($V_{m-1} \rightarrow V_m \leftarrow V_{m+1}$), both arrows point *into* V_m; this is the crucial asymmetric case whose behavior under conditioning is the opposite of the other two.

2.2.2 Blocking Rules

Structure	**Pattern**	**Effect of conditioning**	**Key intuition**
Chain	$A \rightarrow M \rightarrow B$	Blocks the path	Causal transmission
Fork	$A \leftarrow M \rightarrow B$	Blocks the path	Common cause
Collider	$A \rightarrow M \leftarrow B$	**Opens** the path	Common effect

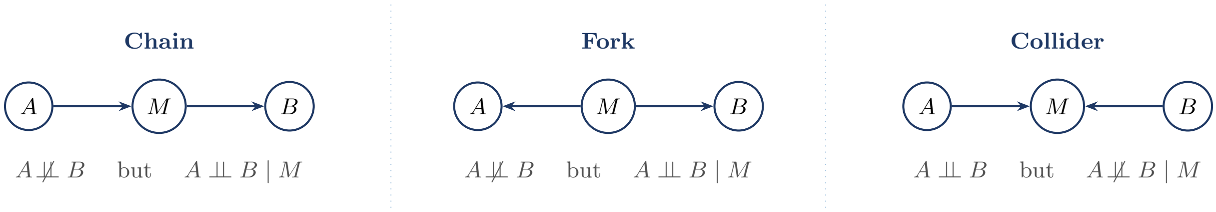


Figure 2.1: The three fundamental path structures in a DAG. Chains and forks are *open by default* and are blocked by conditioning on M. Colliders are *closed by default* and are opened by conditioning on M (or any descendant of M).

The critical asymmetry — colliders are *closed by default* and *opened* by conditioning, while chains and forks are *open by default* and *closed* by conditioning — is what makes collider bias so easy to overlook in practice.

Chain: Smoking → Tar → Cancer. Let A = smoking, M = tar deposits, B = lung cancer. Marginally, heavy smokers have elevated cancer rates — the path is open. Conditioning on tar level blocks the path: once tar level is fixed, the causal channel is fully accounted for.

Fork: Poverty → Poor Diet; Poverty → Lack of Exercise. Let M = poverty, A = poor diet, B = lack of exercise. Unconditionally, diet quality and exercise are correlated through poverty. Conditioning on income level removes this spurious association.

Collider: Accident → Hospitalization ← Cancer. Let A = traffic accident, B = cancer, M = hospitalization. In the general population, $A \perp\!\!\!\perp B$. But among hospitalized patients — conditioning on M — the two become negatively associated. This is Berkson's bias (Berkson 1946).

Three-Node Motifs at a Glance

In a *chain* $A \to M \to B$, conditioning on M blocks the path. In a *fork* $A \leftarrow M \to B$, conditioning on M removes the spurious association. In a *collider* $A \to M \leftarrow B$, the path is blocked by default, but conditioning on M — or on any descendant of M — opens the path and induces association. These three motifs are the local building blocks of d-separation in arbitrary DAGs.

2.2.3 The d-Separation Criterion

Definition: d-Blocking and d-Separation

A path π is *d-blocked* by a set $\mathbf{S}$ if:

1. π contains a chain $A \to M \to B$ or fork $A \leftarrow M \to B$ with $M \in \mathbf{S}$; or
2. π contains a collider $A \to C \leftarrow B$ with $C \notin \mathbf{S}$ and no descendant of C is in $\mathbf{S}$.

Nodes X and Y are *d-separated* by $\mathbf{S}$, written $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$, if every path between X and Y in $\mathcal{G}$ is d-blocked by $\mathbf{S}$.

Example: Applying the d-Separation Definition to the Three Toy Settings

(i) Chain. A = smoking, M = tar, B = cancer. Only path: $A \to M \to B$.

- $\mathbf{S} = \emptyset$: not d-blocked. $\Rightarrow A \not\perp\!\!\!\perp B$.
- $\mathbf{S} = \{M\}$: clause (1) applies. $\Rightarrow A \perp\!\!\!\perp B \mid M$.

(ii) Fork. M = poverty, A = poor diet, B = lack of exercise. Only path: $A \leftarrow M \to B$.

- $\mathbf{S} = \emptyset$: not d-blocked. $\Rightarrow A \not\perp\!\!\!\perp B$.
- $\mathbf{S} = \{M\}$: clause (1) applies. $\Rightarrow A \perp\!\!\!\perp B \mid M$.

(iii) Collider. A = accident, M = hospitalization, B = cancer. Only path: $A \to M \leftarrow B$.

- $\mathbf{S} = \emptyset$: collider $M \notin \mathbf{S}$, clause (2) applies. $\Rightarrow A \perp\!\!\!\perp B$.
- $\mathbf{S} = \{M\}$: clause (2) no longer blocks. $\Rightarrow A \not\perp\!\!\!\perp B \mid M$ (Berkson's bias).

Example: A Direct Probability Proof for the Chain

Consider the chain $X_1 \to X_2 \to X_3$. By the Markov factorization, $p(x_1, x_2, x_3) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)$. Conditioning on $X_2 = x_2$:

$$p(x_1, x_3 \mid x_2) = \frac{p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)}{p(x_2)} = p(x_1 \mid x_2)\, p(x_3 \mid x_2).$$

Hence $X_1 \perp\!\!\!\perp X_3 \mid X_2$. The fork case follows by an identical calculation.

Theorem: Verma & Pearl (1988)

(Pearl 2009, Theorem 1.2.4) If $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$, then $X \perp\!\!\!\perp Y \mid \mathbf{S}$ in every distribution that is Markov with respect to $\mathcal{G}$.

Remark: Soundness, Not Converse

Theorem 2.1 (Verma & Pearl) states that d-separation implies conditional independence for every distribution Markov with respect to $\mathcal{G}$. The converse need not hold without additional assumptions such as *faithfulness*.

Table 2.1 summarizes the Markov factorization and its independence implication for each of the three path structures.

Table 2.1: Markov factorization and d-separation result for each path structure.

Structure	**Markov factorization** $p(a, m, b)$	**d-separation**
Chain $A \to M \to B$	$p(a)\, p(m \mid a)\, p(b \mid m)$	$(A \perp\!\!\!\perp B \mid M)_{\mathcal{G}}$
Fork $A \leftarrow M \to B$	$p(m)\, p(a \mid m)\, p(b \mid m)$	$(A \perp\!\!\!\perp B \mid M)_{\mathcal{G}}$
Collider $A \to M \leftarrow B$	$p(a)\, p(b)\, p(m \mid a, b)$	$(A \perp\!\!\!\perp B)_{\mathcal{G}}$

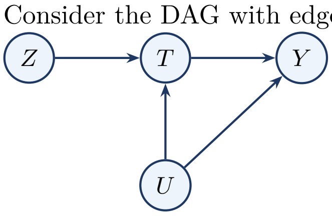
2.2.4 Practical Ways to Check d-Separation

Bayes-Ball Intuition. Imagine releasing a ball from X asking whether it can reach Y, given that nodes in $\mathbf{S}$ are observed. At a chain or fork node not in $\mathbf{S}$: ball passes through. At a chain or fork node in $\mathbf{S}$: ball stops. At a collider not in $\mathbf{S}$ (and no descendant in $\mathbf{S}$): ball stops. At a collider in $\mathbf{S}$ (or with a descendant in $\mathbf{S}$): ball passes through (Shachter 1998).

Moral Graph Transformation. (1) Restrict to the ancestral set $\mathrm{An}(X \cup Y \cup \mathbf{S})$. (2) Moralise: for every collider $A \to C \leftarrow B$, add an undirected edge $A - B$. (3) Remove all edge directions. (4) Delete all nodes in $\mathbf{S}$. If X and Y are disconnected, then $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$ (Lauritzen 1996, Ch. 3).

Example: Checking d-Separation in a Four-Node DAG

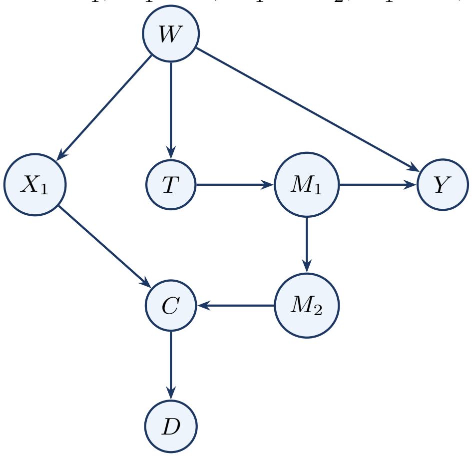
Consider the DAG with edges $Z \to T$, $U \to T$, $T \to Y$, $U \to Y$.



Query 1. Is $(Z \perp\!\!\!\perp U)_{\mathcal{G}}$? The only path $Z \to T \leftarrow U$ has T as a collider with $T \notin \emptyset$ — blocked. $(Z \perp\!\!\!\perp U)_{\mathcal{G}}$.
Query 2. Is $(Z \perp\!\!\!\perp U \mid T)_{\mathcal{G}}$? Now $T \in \mathbf{S}$: the collider is observed, the path is open. $Z \not\!\perp\!\!\!\perp U \mid T$.
Comparing the two queries: Z and U are marginally independent but become dependent once we condition on T. This is Berkson's bias.

Example: Practice DAG — Eight-Node Graph

Consider the DAG with nodes W, X_1, T, M_1, M_2, Y, C, D and edges $W \to X_1$, $W \to T$, $W \to Y$, $T \to M_1$, $M_1 \to Y$, $M_1 \to M_2$, $X_1 \to C$, $M_2 \to C$, $C \to D$.



(1) Is $X_1 \perp\!\!\!\perp Y$? No. Two open paths: $X_1 \leftarrow W \to Y$ and $X_1 \leftarrow W \to T \to M_1 \to Y$.
(2) Is $X_1 \perp\!\!\!\perp Y \mid W$? Yes. Both open paths pass through the fork W; conditioning on W blocks them.
(3) Is $T \perp\!\!\!\perp M_2 \mid M_1$? Yes. The path $T \to M_1 \to M_2$ is blocked by M_1; all other paths are also blocked.
(4) Does conditioning on C create collider bias? Yes. C is a collider on $X_1 \to C \leftarrow M_2$. Conditioning on C opens this path, inducing a spurious association between X_1 and M_2.
(5) Does conditioning on D open the path $X_1 \to C \leftarrow M_2$? Yes. D is a descendant of C. By clause (2) of Definition 2.1, the collider at C is activated.

2.3 Collider Bias and the IV DAG

2.3.1 Berkson's Bias

Definition: Collider Bias

Collider bias is the spurious association induced between two variables when one conditions on their common effect (a collider), or on a descendant of that collider. In a path $A \to C \leftarrow B$, the variables A and B may be marginally independent, but conditioning on C — or on any descendant of C — can make them statistically dependent.

Collider Bias vs. Confounding

Unlike confounding, which is *removed* by conditioning on a common cause, collider bias is *created* by conditioning on a common effect. This is why conditioning on more variables is not always safer in causal analysis: adding a collider or a descendant of a collider to the adjustment set introduces bias rather than reducing it.

Example: Collider Bias in the IV DAG

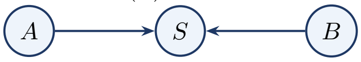
In the IV DAG, T is a collider on the path $Z \to T \leftarrow U$. Marginally, $Z \perp\!\!\!\perp U$ (the instrument is exogenous). However,

$$Z \perp\!\!\!\perp U \qquad \text{but} \qquad Z \not\!\perp\!\!\!\perp U \mid T.$$

Conditioning on T — restricting to the treated subpopulation — makes the instrument Z appear associated with U, invalidating the IV argument within that stratum.

Example: Collider Bias through Selection: Talent and Wealth

Suppose both talent (A) and wealth (B) increase the probability of admission to an elite school, so admission (S) is a collider: $A \to S \leftarrow B$.



In the general population, $A \perp\!\!\!\perp B$. But among admitted students — conditioning on S — lower talent makes higher wealth more likely, and vice versa. This is collider bias: restricting to admitted students is precisely conditioning on a common effect.

2.3.2 Full d-Separation Analysis of the IV DAG

We work through the IV DAG with edges $Z \to T$, $T \to Y$, $U \to T$, $U \to Y$, where U is unobserved.

Endpoints	**Path**	**Type**	**Why**
$Z \leftrightarrow Y$	$Z \to T \to Y$	*Causal*	Directed from Z to Y; carries the IV signal
$Z \leftrightarrow Y$	$Z \to T \leftarrow U \to Y$	*Associational*	Collider at T; activates back-door via U
$Z \leftrightarrow U$	$Z \to T \leftarrow U$	*Associational*	Collider at T; dormant unless T conditioned on

Question 1. Is $Z \perp\!\!\!\perp Y$? There are two paths. The directed path $Z \to T \to Y$ is open, since it is a chain and we are not conditioning on T. The path $Z \to T \leftarrow U \to Y$ contains a collider at T; since we are not conditioning on T, that path is blocked. Because the causal path remains open, $Z \not\perp\!\!\!\perp Y$.

Question 2. Is $Z \perp\!\!\!\perp Y \mid T$? The causal path $Z \to T \to Y$ is blocked because conditioning on T blocks the chain. The path $Z \to T \leftarrow U \to Y$ contains a collider at T, and conditioning on T opens it. Hence $Z \not\perp\!\!\!\perp Y \mid T$. The residual association is entirely spurious — a collider artefact, not a causal effect.

Question 3. Is $Z \perp\!\!\!\perp Y \mid \{T, U\}$? The causal path is blocked by T. The path $Z \to T \leftarrow U \to Y$ has its collider at T opened by conditioning on T, but then blocked at U because we also condition on U. Hence $Z \perp\!\!\!\perp Y \mid \{T, U\}$.

Question 4. Is $Z \perp\!\!\!\perp U$? The only path $Z \to T \leftarrow U$ contains a collider at T. Since we are not conditioning on T or any descendant of T, the path is blocked. Hence $Z \perp\!\!\!\perp U$. This is *exogeneity*.

The four questions give a complete graphical account of the three IV assumptions. Question 1 establishes *relevance*; Question 4 establishes *exogeneity*; Question 3 establishes the *exclusion restriction*. Question 2 is the warning: conditioning on T alone simultaneously closes the causal channel and opens the confounded channel.

2.4 The Markov Property and Factorization

Proposition: Conditional Independence in the Three-Node Chain

Let $X_1 \to X_2 \to X_3$ be a chain with Markov factorization $p(x_1, x_2, x_3) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)$. Then $X_1 \perp\!\!\!\perp X_3 \mid X_2$. The analogous result for the fork $X_1 \leftarrow X_2 \to X_3$ is left as an exercise.

Definition: Local Markov Property

A distribution P satisfies the *local Markov property* with respect to $\mathcal{G}$ if, for every node $V_i \in \mathcal{V}$:

$$V_i \perp\!\!\!\perp (\mathrm{Nd}(V_i)\ \ \mathrm{Pa}(V_i)) \mid \mathrm{Pa}(V_i).$$

That is, once we condition on the direct causes of a node, that node is independent of all variables that are neither its descendants nor its parents.

Remark

The *global Markov property* is the statement that every d-separation in $\mathcal{G}$ implies a conditional independence in P — precisely the content of Theorem 2.1 (Verma & Pearl). For DAGs, the local and global Markov properties are equivalent (Lauritzen 1996, Proposition 3.27).

An immediate consequence is the *Markov factorization*:

$$p(v_1, \dots, v_k) \;=\; \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i))\,. \tag{2.1}$$

Example: Markov Factorization in the Education Example

For the DAG with edges $N \to E$, $B \to E$, $B \to Y$, $E \to Y$:

$$p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b).$$

A regression of Y on E alone is confounded by B. Conditioning on B blocks the back-door path $E \leftarrow B \to Y$ and identifies $P(y \mid \mathrm{do}(E{=}e))$ via the back-door formula (Chapter 3).

Example: Markov Factorization in the IV DAG

Including the latent U: $p(z, t, y, u) = p(z)\, p(u)\, p(t \mid z, u)\, p(y \mid t, u)$. Since U is unobserved, the observable factorization is obtained by marginalizing:

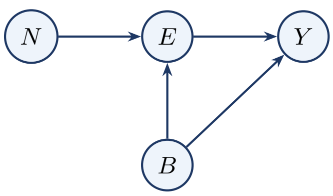
$$p(z, t, y) = \int p(z)\, p(u)\, p(t \mid z, u)\, p(y \mid t, u)\, \mathrm{d}u.$$

This cannot be simplified to $p(z)\, p(t \mid z)\, p(y \mid t)$ when U is a common cause of T and Y: this is precisely the endogeneity problem.

2.5 Worked Example: The Education–Earnings DAG

This section is the template for how we will use DAGs throughout the course: specify the DAG, read off the factorization, use d-separation to identify implied conditional independences, and interpret the result in causal terms.

The causal story. Education (E) affects earnings (Y); family background (B) is a common cause of both; neighborhood (N) affects education but has no direct effect on earnings. All four variables are observed; there are no latent confounders.



Step 1 — Markov factorization. $\mathrm{Pa}(N) = \mathrm{Pa}(B) = \emptyset$, $\mathrm{Pa}(E) = \{N, B\}$, $\mathrm{Pa}(Y) = \{E, B\}$.

$$p(n, b, e, y) \;=\; p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b).$$

Step 2 — d-Separation. Two paths between N and Y: Path 1: $N \to E \to Y$ (chain). Path 2: $N \to E \leftarrow B \to Y$ (collider at E).

- *Query (i):* $(N \perp\!\!\!\perp Y)_{\mathcal{G}}$? Path 1 is a chain with nothing conditioned on — open. $N \not\perp\!\!\!\perp Y$.
- *Query (ii):* $(N \perp\!\!\!\perp Y \mid E)_{\mathcal{G}}$? Path 1 blocked by E; Path 2 collider E opened, path continues through B which is open. $N \not\perp\!\!\!\perp Y \mid E$. This is a collider trap: conditioning on E opens a spurious association between N and Y through B.

- *Query (iii):* $(N \perp\!\!\!\perp Y \mid \{E, B\})_{\mathcal{G}}$? Path 1 blocked by E; Path 2 opened at E but blocked at B. $(N \perp\!\!\!\perp Y \mid E, B)_{\mathcal{G}}$.
- *Query (iv):* $(N \perp\!\!\!\perp B)_{\mathcal{G}}$? Only path $N \to E \leftarrow B$ has collider E with $E \notin \emptyset$ — blocked. $N \perp\!\!\!\perp B$.

Step 3 — Conditional Independence. By Theorem 2.1 (Verma & Pearl): $N \perp\!\!\!\perp B$, $N \perp\!\!\!\perp Y \mid E, B$, $N \not\perp\!\!\!\perp Y$, $N \not\perp\!\!\!\perp Y \mid E$.

Step 4 — Identification. The only back-door path is $E \leftarrow B \to Y$. With $\mathbf{S} = \{B\}$:

$$P(y \mid \mathrm{do}(E{=}e)) \;=\; \sum_b P(y \mid e, b)\, P(b).$$

2.6 The Big Picture

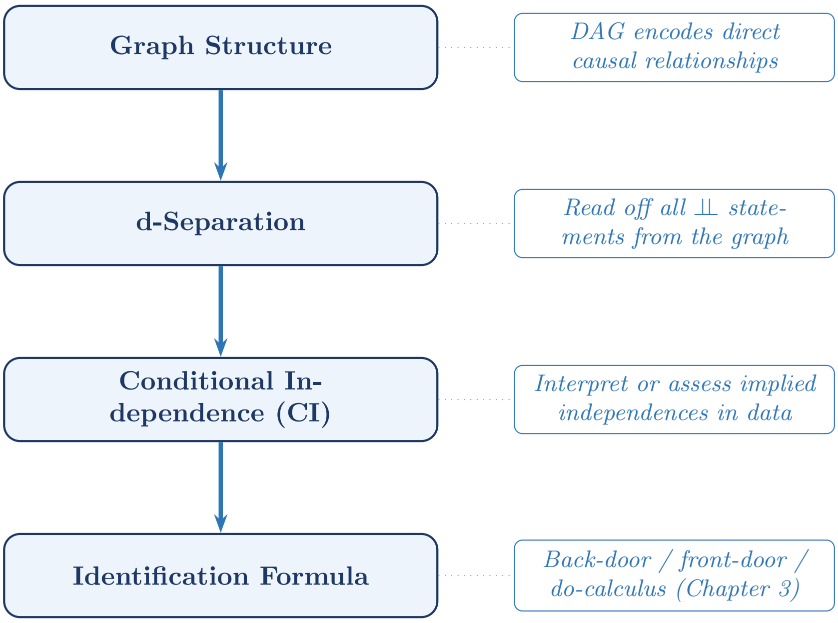


Figure 2.2: The four-step pipeline from causal assumptions to identification formula.

The logical chain in compact form:

structural assumptions $\implies$ DAG $\implies$ d-separation $\implies$ CI under the Markov property $\implies$ identification formulas.

2.7 Summary

1. **DAGs as causal structure.** A DAG is a directed acyclic graph whose nodes represent variables and whose directed edges represent direct causal relationships. Once a DAG is specified, it encodes which variables are causally connected, which are potential confounders, and which may block or open paths.

2. **Three-node motifs.** Every path is built from three local structures: chains, forks, and colliders. Chains and forks are open by default and blocked by conditioning on the middle node. Colliders are blocked by default and opened by conditioning on the collider or one of its descendants.

3. **d-Separation and conditional independence.** The d-separation criterion is a graphical rule: X and Y are d-separated by $\mathbf{S}$ exactly when every path between them is blocked by $\mathbf{S}$. Under the Markov property, this implies the corresponding conditional independence. Some such implications may be assessed empirically, but agreement with data does not by itself verify the graph.

4. **Markov factorization.** The local Markov property yields

$$p(v_1, \ldots, v_k) \;=\; \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i))\,,$$

which expresses the joint distribution in terms of local conditional distributions.

5. **Collider bias.** Conditioning on a collider — or on a descendant of a collider — can induce a spurious association between otherwise independent variables (Berkson 1946). This is the *opposite* of confounding adjustment: conditioning on a common cause removes spurious association, whereas conditioning on a common effect creates it.

6. **A practical workflow.** To analyze a DAG, proceed in four steps: identify the graph structure, determine which paths are open or blocked, translate d-separation statements into conditional independences under the Markov property, and interpret those independences in light of the causal question.

2.8 Problems

1. Warm-up: a single collider. Consider the DAG $X \to Y \leftarrow Z$, where X and Z have no other connections.

(a) Identify the structural role of Y on the path $X \to Y \leftarrow Z$.
(b) Is $X \perp\!\!\!\perp Z$? Apply the d-separation criterion and state which blocking rule applies.
(c) Is $X \perp\!\!\!\perp Z \mid Y$? Explain what happens to the path when Y is conditioned on.

2. d-Separation practice. Consider the DAG: $A \to B \to D$, $A \to C \to D$, $B \to E$, $C \to E$.

(a) List all paths between A and E. (*Hint*: there are four paths; two pass through D.)
(b) For each path, identify the role of each intermediate node.
(c) Does $\{B, C\}$ d-separate A and E?
(d) Does $\{D\}$ d-separate B and C? What type of node is D on the path $B \to D \leftarrow C$?

3. Berkson's bias. Suppose X and Y are independent standard normal variables, and let $S = \mathbf{1}[X+Y > 0]$ (selected into a sample).

(a) Verify analytically that $\mathrm{Cov}(X, Y \mid S{=}1) < 0$.
(b) Draw the DAG for (X, Y, S) and identify S as a collider.
(c) Explain in one sentence why restricting the analysis to the subsample with $S = 1$ biases estimates of any association between X and Y, and name the type of bias this illustrates.

4. Markov factorization and collider activation. Consider the DAG with edges $A \to E$, $A \to W$, $F \to E$, $E \to W$, where A = ability, F = family income, E = education, W = wages.

(a) Write down the Markov factorization $p(a, f, e, w)$.
(b) Is $(F \perp\!\!\!\perp W)_{\mathcal{G}}$? List all paths between F and W.
(c) Is $(F \perp\!\!\!\perp W \mid E)_{\mathcal{G}}$? Identify the role of E on each path.
(d) A researcher regresses W on E and F, omitting A. Is the coefficient on E a causal effect?

5. Theorem for the collider. Consider the collider $A \to M \leftarrow B$ with Markov factorization $p(a, m, b) = p(a)\, p(b)\, p(m \mid a, b)$.

(a) By marginalizing over M, show that $A \perp\!\!\!\perp B$ in the joint distribution.
(b) Show that conditioning on M breaks this independence.
(c) Explain why parts (a) and (b) together are consistent with Theorem 2.1 (Verma & Pearl).

6. Terminology check. Consider the DAG with edges $U \to X$, $U \to Z$, $X \to W$, $Z \to W$, $W \to Y$.

(a) Identify the parents, children, ancestors, descendants, and non-descendants of node W.
(b) List all pairs of adjacent nodes.
(c) Which pairs of nodes are connected by a directed path?
(d) Write the Markov factorization $p(u, x, z, w, y)$.

7. Markov factorization and local Markov property. Consider the DAG with edges $X_1 \to X_2$, $X_1 \to X_3$, $X_2 \to X_4$, $X_3 \to X_4$.

(a) Write the joint density $p(x_1, x_2, x_3, x_4)$ implied by the Markov factorization.
(b) State the local Markov property for each of the four nodes.
(c) Is $(X_2 \perp\!\!\!\perp X_3)_{\mathcal{G}}$? Is $(X_2 \perp\!\!\!\perp X_3 \mid X_1)_{\mathcal{G}}$?

8. Toy proof: conditional independence in the fork. Consider the fork $X_1 \leftarrow X_2 \to X_3$ with Markov factorization $p(x_1, x_2, x_3) = p(x_2)\, p(x_1 \mid x_2)\, p(x_3 \mid x_2)$.

(a) Show directly that $X_1 \perp\!\!\!\perp X_3 \mid X_2$.
(b) Is $X_1 \perp\!\!\!\perp X_3$ marginally? Justify both graphically and algebraically.
(c) Explain why conditioning on the common cause X_2 removes the association.

9. d-Separation in the practice DAG. Refer to the eight-node DAG in the eight-node example above. For each query, state whether it is true or false and justify by listing all relevant paths.

(a) $X_1 \perp\!\!\!\perp Y$
(b) $X_1 \perp\!\!\!\perp Y \mid W$
(c) $T \perp\!\!\!\perp M_2 \mid M_1$
(d) Does conditioning on C induce collider bias between X_1 and M_2? Identify the specific path that is opened and explain why the induced association is spurious.
(e) Does conditioning on D (a descendant of C) open the path $X_1 \to C \leftarrow M_2$? State which clause of Definition 2.1 applies.

Chapter 3

The Do-Calculus and Identification Criteria

Learning Objectives

By the end of this chapter, students should be able to:

1. Define the two intervention graph operations $\mathcal{G}_{\overline{X}}$ and $\mathcal{G}_{\underline{X}}$, explain the causal meaning of $\mathcal{G}_{\overline{X}}$, and describe the technical role of $\mathcal{G}_{\underline{X}}$.
2. Apply the back-door criterion to determine whether a set $\mathbf{S}$ identifies a causal effect, and write the back-door adjustment formula.
3. Apply the front-door criterion in graphs where the confounder is unobserved but a suitable mediator exists.
4. State all three rules of do-calculus, identify the graph condition that licenses each rule, and use them to prove the back-door and front-door formulas.
5. State the completeness theorem of Shpitser and Pearl (2006) and explain its practical implication: if the do-calculus cannot identify a quantity from the observational distribution relative to the assumed graph, then no purely observational method can do so without additional assumptions or new data.

Remark: How to Read This Chapter

This chapter has two pedagogical layers. Sections 1–3 develop the two most important concrete identification criteria — back-door and front-door adjustment — using intervention graphs and direct graphical reasoning. These sections are the core material for a first reading.
Sections 4–5 then place these criteria inside the more general theory of the do-calculus and identifiability. These later sections are conceptually important, but more abstract and may be read as a second pass after the concrete criteria are understood.

3.1 From d-Separation to Intervention: Intervention Graphs

In Chapter 2 we learned to read conditional independence from a DAG using d-separation. This chapter takes the next step: translating the graphical language into *identification formulas* — expressions that write the interventional distribution $P(y \mid \mathrm{do}(T{=}t))$ entirely in terms of quantities observable from data.

3.1.1 What Does Intervention Mean?

Conditioning versus intervening. The conditional distribution $P(Y \mid T{=}t)$ describes the subpopulation of units for whom T was *observed* to equal t. Because treatment assignment may be influenced by confounders, this subpopulation is not representative of the full population. The interventional distribution $P(Y \mid \mathrm{do}(T{=}t))$, by contrast, describes the population that would result if T were *externally set* to t for everyone — removing the natural process that determines T and replacing it with the forced value. The two distributions coincide only when T has no unmeasured common causes with Y.

A simple illustration: suppose temperature Z causes both ice-cream sales T and crime Y. The quantity $P(Y \mid T{=}t)$ is the crime rate on days when ice-cream sales happen to equal t — confounded by Z. The quantity $P(Y \mid \mathrm{do}(T{=}t))$ is the crime rate we would observe if we *fixed* sales at level t by decree. Forcing sales does not change the temperature, so crime remains driven only by Z, not by the sales level we mandated.

The structural basis of the do-operator. In the SEM framework of Chapter 1, an intervention $\mathrm{do}(T{=}t)$ **replaces the entire structural equation** for T with the constant $T = t$. The replacement severs T's dependence on its parents $\mathrm{Pa}(T)$, while all other structural equations remain unchanged.

Graph surgery as the graphical realization. Because $\mathrm{Pa}(T)$ no longer affects T after the intervention, every arrow pointing *into* T in the DAG is removed. The resulting graph — the *intervention graph* $\mathcal{G}_{\overline{T}}$ — represents the post-intervention world. D-separation in $\mathcal{G}_{\overline{T}}$ encodes conditional independence in $P(\cdot \mid \mathrm{do}(T{=}t))$, not in the original observational distribution P.

3.1.2 The Two Graph Operations

Definition: Intervention Graphs $\mathcal{G}_{\overline{X}}$ and $\mathcal{G}_{\underline{X}}$

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a DAG and $X \subseteq \mathcal{V}$ a set of nodes.

- $\mathcal{G}_{\overline{X}}$ (*intervention graph*, or *mutilated graph*) is obtained by *deleting all arrows into* X. This represents the intervention $\mathrm{do}(X{=}x)$, which replaces the structural equation for X by the constant x, severing X's dependence on all its former parents.
- $\mathcal{G}_{\underline{X}}$ (*auxiliary observation graph*) is obtained by *deleting all arrows out of* X. This is a *technical device* used in graphical conditions for certain do-calculus steps. It does *not* represent a probabilistic model of conditioning on X; it is simply the graph in which the relevant d-separation condition must be checked.

Two Graph Operations at a Glance

- **Delete arrows *into*** X ($\mathcal{G}_{\overline{X}}$): models the intervention $\mathrm{do}(X{=}x)$. D-separation in $\mathcal{G}_{\overline{X}}$ encodes independence in the post-intervention distribution $P(\cdot \mid \mathrm{do}(X{=}x))$.
- **Delete arrows *out of*** X ($\mathcal{G}_{\underline{X}}$): a technical device used in graphical conditions for certain do-calculus steps. It is *not* a model of conditioning on X.

Figure 3.1 illustrates both graph operations on the IV DAG $\{Z \to T,\ T \to Y,\ U \to T,\ U \to Y\}$.

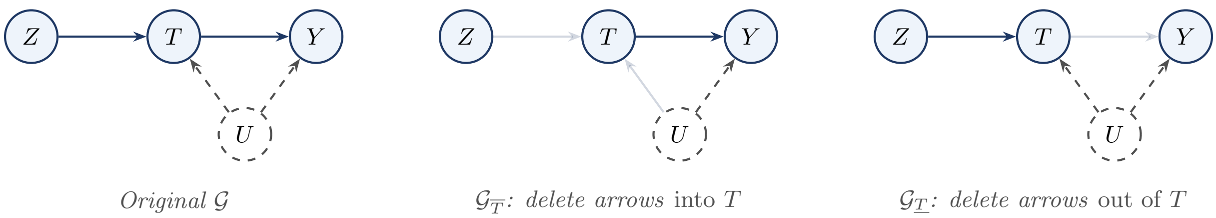


Figure 3.1: The two graph operations on the IV DAG. Faded arrows have been removed. $\mathcal{G}_{\overline{T}}$ models external intervention $\mathrm{do}(T{=}t)$: the back-door path $T \leftarrow U \to Y$ is severed. $\mathcal{G}_{\underline{T}}$ is a technical device used in graphical conditions for certain do-calculus steps; it does not represent conditioning on T.

3.1.3 Why the Intervention Graph Works

The Intervention Principle

In the original DAG $\mathcal{G}$, the node T is determined by its parents $\mathrm{Pa}(T)$ via the structural equation $T = f_T(\mathrm{Pa}(T), \varepsilon_T)$. Intervening on T removes the structural mechanism that normally determines T; therefore the parents of T no longer influence it. The intervention $\mathrm{do}(T{=}t)$ *replaces this equation* with the constant $T = t$, making T independent of all its former parents. Graphically, this severs every arrow into T. The remaining structure of the graph is unchanged: all other structural equations still hold.

A d-separation statement in $\mathcal{G}_{\overline{T}}$ is therefore a conditional independence statement in the *post-intervention* distribution $P(\cdot \mid \mathrm{do}(T{=}t))$, not in the observational distribution P.

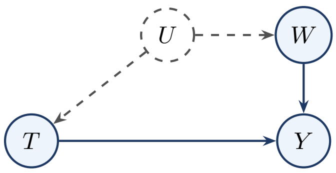
3.2 The Back-Door Criterion

The back-door criterion is the most widely used identification strategy. It gives a simple graphical condition under which conditioning on an observed set $\mathbf{S}$ is sufficient to identify the causal effect of T on Y.

3.2.1 Motivation: When the Confounder Is Unobserved

If all common causes of T and Y were observed, identification would be trivial: adjust for all of them. The interesting case is when some confounders are *unobserved*, yet an observed set $\mathbf{S}$ can still block every spurious path.

Consider the following DAG: a drug T affects recovery Y. Socioeconomic status U is an unobserved confounder ($U \to T$, $U \to Y$), but U also affects an observed proxy W ($U \to W$), which in turn affects Y ($W \to Y$).



The only back-door path is $T \leftarrow U \to W \to Y$. Although U is unobserved, W sits on this path and *is* observed. Conditioning on W blocks the path at the chain node W, and W is not a descendant of T, so $\mathbf{S} = \{W\}$ satisfies the back-door criterion. The causal effect is identified even though U is never measured. The criterion does not require adjusting for the confounders themselves, only for an observed set that *intercepts* every back-door path.

3.2.2 Definition and Theorem

Definition: Back-Door Criterion

A set of observed variables $\mathbf{S}$ satisfies the *back-door criterion* for the causal effect of T on Y in $\mathcal{G}$ if:

1. No node in $\mathbf{S}$ is a descendant of T.
2. $\mathbf{S}$ blocks every *back-door path* from T to Y, i.e., every path that begins with an arrow pointing *into* T.

Remark

The terminology reflects the geometry of the graph relative to T. A *back-door path* enters T from behind — it represents a confounding route through which T and Y become spuriously associated without any causal mechanism from T to Y. The front-door criterion of Section 3.3 is named for the complementary idea: it identifies the causal effect by intercepting the *front-door* (causal) paths at a mediator M, rather than blocking the back-door (confounding) paths directly. Standing at T, confounders sneak in through the back door; the causal effect walks out through the front.

Remark

Condition 2 is equivalent to requiring $(Y \perp\!\!\!\perp T \mid \mathbf{S})_{\mathcal{G}_{\underline{T}}}$: $\mathbf{S}$ d-separates T and Y in the graph $\mathcal{G}_{\underline{T}}$ obtained by *deleting all arrows out of* T. Removing T's outgoing edges eliminates the causal paths $T \to \cdots \to Y$, leaving only back-door paths between T and Y.
The back-door criterion adds two constraints: $\mathbf{S}$ must consist of *observed* variables only, and it must contain no descendant of T. Every valid adjustment set is therefore a valid d-separator in $\mathcal{G}_{\underline{T}}$, but the converse does not hold.

Example: A Single Confounder

Consider the DAG $T \leftarrow C \to Y$ with $T \to Y$, where C is an observed confounder. There is one back-door path: $T \leftarrow C \to Y$.
Does $\mathbf{S} = \{C\}$ satisfy the back-door criterion?

1. C is not a descendant of T.
2. $\{C\}$ blocks $T \leftarrow C \to Y$ (fork at C).

The back-door formula gives: $P(y \mid \mathrm{do}(T{=}t)) = \sum_c P(y \mid t, c)\, P(c)$.

Example: Two Back-Door Paths — Both Must Be Blocked

Consider treatment T, outcome Y, observed variables C_1 and C_2, and an unobserved confounder U: $C_1 \to T$, $C_1 \to Y$, $U \to T$, $U \to C_2$, $C_2 \to Y$, $T \to Y$.
There are exactly two back-door paths from T to Y:

1. $T \leftarrow C_1 \to Y$ (direct confounding by the observed C_1),
2. $T \leftarrow U \to C_2 \to Y$ (indirect confounding via the unobserved U, routing through the observed proxy C_2).

$\mathbf{S} = \{C_1\}$ fails: it blocks path 1 but leaves path 2 open. $\mathbf{S} = \{C_2\}$ fails: it blocks path 2 but leaves path 1 open. $\mathbf{S} = \{C_1, C_2\}$ succeeds: it blocks both paths, and neither C_1 nor C_2 is a descendant of T.

M-Bias: More Variables Can Be Worse

Consider a graph where T and Y each have their own separate unobserved confounders, U_1 and U_2, that happen to share a common observed descendant C: $U_1 \to T$, $U_2 \to Y$, $U_1 \to C$, $U_2 \to C$, $T \to Y$ (with U_1, U_2 unobserved).
Marginally, the only back-door path $T \leftarrow U_1 \to C \leftarrow U_2 \to Y$ is *blocked* by the collider C. The causal effect is already identified without any adjustment!
Conditioning on C *activates* the collider, opening the path. The set $\{C\}$ therefore fails Condition 2: it does not block the back-door path; it creates one.
This phenomenon is called *M-bias* because the graph has an M-shaped skeleton. The practical lesson is that Condition 1 alone ("include all pretreatment variables") is not sufficient; every candidate adjustment variable must also be verified not to open a previously closed back-door path.

Theorem: Back-Door Adjustment Formula

(Pearl 1993) If $\mathbf{S}$ satisfies the back-door criterion for the effect of T on Y in $\mathcal{G}$, and $P(T{=}t \mid \mathbf{S}{=}\mathbf{s}) > 0$ for all $\mathbf{s}$ with $P(\mathbf{S}{=}\mathbf{s}) > 0$, then:

$$P(y \mid \mathrm{do}(T{=}t)) \;=\; \int P(y \mid T{=}t,\; \mathbf{S}{=}\mathbf{s})\, dP(\mathbf{s}). \tag{3.1}$$

Equivalently, for any measurable function h:

$$\mathbb{E}[h(Y) \mid \mathrm{do}(T{=}t)] \;=\; \mathbb{E}_{\mathbf{S}}\Big[\mathbb{E}[h(Y) \mid T{=}t,\; \mathbf{S}]\Big]. \tag{3.2}$$

Remark

The positivity condition requires every value of the treatment $T{=}t$ to be observable within each stratum of $\mathbf{S}$: if some stratum $\mathbf{s}$ never receives treatment t in the data, the conditional distribution $P(y \mid t, \mathbf{s})$ is undefined and Equation 3.1 cannot be computed. Positivity is a *data-support* condition, not a graphical one. When positivity fails on a set of measure zero, the formula still identifies $P(y \mid \mathrm{do}(t))$ because the offending strata contribute zero mass to the outer integral.

3.2.3 The Adjustment Formula: Standardization

Equation 3.1 is also known as the *standardization formula* or, in the epidemiological literature, the *g-formula* (Robins 1986). The interpretation is important:

- $P(y \mid t, \mathbf{s})$ is the stratum-specific conditional distribution of Y given $T{=}t$ and $\mathbf{S}{=}\mathbf{s}$. Within each stratum, T is not confounded by any back-door path (all are blocked by $\mathbf{S}$), so this conditional distribution *is* causal.
- The outer integration $\int \cdots \, dP(\mathbf{s})$ weights the strata by the *population marginal* distribution of $\mathbf{S}$, rather than the treatment-conditional distribution $dP(\mathbf{s} \mid t)$ that a naïve analysis would use.
- In a randomized experiment, $\mathbf{S} = \emptyset$ satisfies the back-door criterion (no confounding), and Equation 3.1 reduces to $P(y \mid \mathrm{do}(t)) = P(y \mid t)$: the observational and interventional distributions coincide.

Example: Back-Door Adjustment — A Numerical Walkthrough

Use the single-confounder graph ($T \leftarrow C \rightarrow Y$, $T \rightarrow Y$) with all variables binary: C = disease severity, T = treatment, Y = recovery. Severity is equally prevalent, $P(C{=}1) = 0.5$, and sicker patients are preferentially treated.

C	$P(C)$	$P(T{=}1 \mid C)$	$P(Y{=}1 \mid T{=}0, C)$	$P(Y{=}1 \mid T{=}1, C)$
0	0.50	0.20	0.10	0.40
1	0.50	0.80	0.50	0.80

The true ATE within each stratum is 0.30.
Naïve estimand (ignoring confounding): $P(Y{=}1 \mid T{=}1) = 0.72$, $P(Y{=}1 \mid T{=}0) = 0.18$. Naïve difference: **0.54**.
Back-door adjusted estimand:

$$P(Y{=}1 \mid \mathrm{do}(T{=}1)) = 0.40 \times 0.5 + 0.80 \times 0.5 = 0.60,$$

$$P(Y{=}1 \mid \mathrm{do}(T{=}0)) = 0.10 \times 0.5 + 0.50 \times 0.5 = 0.30.$$

Adjusted difference: **0.30** — matching the true ATE.
The naïve estimate (0.54) overstates the causal effect by 0.24 because treated patients are predominantly severe ($C{=}1$), who recover at higher rates *even without treatment*. The back-door formula removes this confounding by imposing the *same* covariate distribution $P(C)$ on both potential worlds.

3.2.4 Application to the Education–Earnings DAG

Example: Back-Door Identification in the Education Example

This is the Education–Earnings DAG analyzed in full in Chapter 2 (Section 5). The DAG has edges $N \rightarrow E$, $B \rightarrow E$, $B \rightarrow Y$, $E \rightarrow Y$, where N = neighborhood, B = family background, E = education, Y = earnings. We wish to identify $P(y \mid \mathrm{do}(E{=}e))$.
The only back-door path is $E \leftarrow B \rightarrow Y$.

- Does $\{B\}$ satisfy the criterion? B is not a descendant of E ; $\{B\}$ blocks $E \leftarrow B \rightarrow Y$. By

Equation 3.1: $P(y \mid \mathrm{do}(E{=}e)) = \sum_b P(y \mid e, b)\, P(b)$.
- Does $\{N\}$ satisfy the criterion? $\{N\}$ does *not* block $E \leftarrow B \rightarrow Y$ (the fork at B remains open). Fails condition 2.
- Does $\{N, B\}$ satisfy the criterion? Yes: neither N nor B is a descendant of E, and $\{N, B\}$ blocks the only back-door path.

Remark

The Education example illustrates three important points.
(a) Multiple valid sets may exist, and they identify the same quantity. Both $\{B\}$ and $\{N, B\}$ satisfy the back-door criterion here. Any such set yields the same interventional distribution; the identification target is unique even though the adjustment sets are not.
(b) Supersets of a valid adjustment set are not automatically valid. Adding a variable to a valid set can violate Condition 2 if that variable is a collider on some back-door path. In this graph $\{N, B\}$ happens to remain valid, but this must be verified from the graph, not assumed.
(c) Smaller valid sets are often preferable in practice. Adjusting for fewer variables — the minimal sufficient set — is generally more efficient and more robust, because every variable added to $\mathbf{S}$ must also satisfy positivity.

3.3 The Front-Door Criterion

The back-door criterion requires that all confounders be observed. The front-door criterion provides identification in a different configuration: when there exists a measured *mediator* M on the causal path from T to Y with specific independence properties.

3.3.1 The Configuration and Criterion

Definition: Front-Door Criterion

A set of variables $\mathbf{M}$ satisfies the *front-door criterion* for the effect of T on Y in $\mathcal{G}$ if:
1. $\mathbf{M}$ intercepts all directed paths from T to Y (all causal paths pass through $\mathbf{M}$).
2. There are no unblocked back-door paths from T to $\mathbf{M}$ (the effect of T on $\mathbf{M}$ is not confounded).
3. All back-door paths from $\mathbf{M}$ to Y are blocked by T (conditioning on T closes all confounding of the $\mathbf{M} \rightarrow Y$ effect).

Example: A Single Mediator with Unobserved Confounding

Consider the DAG $T \rightarrow M \rightarrow Y$ with $U \rightarrow T$ and $U \rightarrow Y$, where M is observed and U is unobserved. Does M satisfy the front-door criterion?
1. *M intercepts all directed paths from T to Y.* The only directed path is $T \rightarrow M \rightarrow Y$, which passes through M.
2. *No unblocked back-door paths from T to M.* There is no edge $U \rightarrow M$, so no back-door path exists.
3. *All back-door paths from M to Y are blocked by T.* The only back-door path is $M \leftarrow T \leftarrow U \rightarrow Y$. Conditioning on T blocks this path (chain at T).

All three conditions hold, so the front-door formula applies:

$$P(y \mid \mathrm{do}(T{=}t)) = \sum_m P(m \mid t) \sum_{t'} P(y \mid t', m)\, P(t').$$

Example: When the Front-Door Criterion Fails

Each condition of the definition is load-bearing.

Case (a): Condition 2 fails — the $T \to M$ link is confounded. Add the edge $U \to M$ so the unobserved confounder also directly affects the mediator. Condition 1 still holds, but condition 2 now fails: the path $T \leftarrow U \to M$ is an unblocked back-door path from T to M. Stage 1 of the front-door formula uses $P(M{=}m \mid T{=}t)$ as if the $T \to M$ link were unconfounded. But with $U \to M$, the association between T and M is partly due to their shared cause U, not purely the causal effect $T \to M$. Stage 1 no longer estimates $P(m \mid \mathrm{do}(T{=}t))$.

Case (b): Condition 3 fails — the $M \to Y$ link has an extra unobserved confounder. Add a second unobserved variable V with $V \to M$ and $V \to Y$. Conditions 1 and 2 still hold. But condition 3 now fails: the path $M \leftarrow V \to Y$ is a back-door path from M to Y that is *not* blocked by conditioning on T. Stage 2 uses $\sum_{t'} P(y \mid t', m)\, P(t')$ as the back-door-adjusted effect of M on Y, but T is no longer a valid adjustment set for Stage 2.

Modification	**Condition violated**	**Stage broken**
Add $U \to M$	Cond. 2: $T \to M$ confounded	Stage 1
Add $V \to M$ and $V \to Y$	Cond. 3: $M \to Y$ confounded by V	Stage 2

The prototypical graph where the front-door criterion applies is: $T \to M \to Y$, $U \to T$, $U \to Y$, no direct $T \to Y$ edge. Figure 3.2 shows this graph.

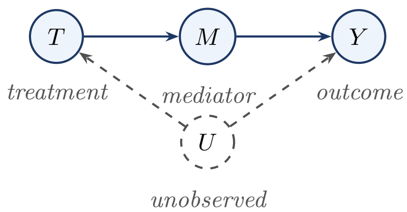


Figure 3.2: The prototypical front-door graph. The confounder U is unobserved, so back-door adjustment on U is infeasible. However, the mediator M is fully observed and satisfies the three front-door conditions, enabling identification via the front-door formula.

3.3.2 The Front-Door Formula

The front-door strategy splits the total effect of T on Y into two identifiable pieces: the effect of T on the mediator M, and the effect of M on Y. Front-door identification does not remove confounding directly; it *routes around it* through an observed causal pathway.

Theorem: Front-Door Adjustment Formula

If M satisfies the front-door criterion for the effect of T on Y (Pearl 1995), and $P(t) > 0$ for all t, then:

$$P(y \mid \mathrm{do}(T{=}t)) \;=\; \sum_m \underbrace{P(m \mid T{=}t)}_{\text{Stage 1}} \; \underbrace{\sum_{t'} P(y \mid T{=}t',\; M{=}m)\, P(t')}_{\text{Stage 2}}. \tag{3.3}$$

Equivalently, writing $\mu(t', m) = \mathbb{E}[Y \mid T{=}t',\; M{=}m]$:

$$\mathbb{E}[Y \mid \mathrm{do}(T{=}t)] \;=\; \mathbb{E}_{M|T=t}\Big[\mathbb{E}_T[\mu(T,\, M)]\Big]. \tag{3.4}$$

The formula chains two stages: **Stage 1** $P(m \mid T{=}t)$ is the distribution of the mediator given $T{=}t$. Condition 2 guarantees no confounding between T and M, so this is directly identifiable. **Stage 2** $\sum_{t'} P(y \mid T{=}t',\; M{=}m)\, P(t')$ is the back-door-adjusted causal effect of M on Y, with T as the adjustment variable. Condition 3 guarantees that conditioning on T blocks all back-door paths from M to Y.

Derivation Sketch

The formula is the composition of two back-door applications.

1. **Expand over the mediator.** By condition 1, every directed path from T to Y passes through M:
$$P(y \mid \mathrm{do}(T{=}t)) = \sum_m P(y \mid \mathrm{do}(M{=}m))\, P(m \mid \mathrm{do}(T{=}t)).$$
2. **Identify the distribution of M under intervention.** By condition 2 there are no back-door paths from T to M, so T satisfies the (trivial) back-door criterion for its effect on M with empty adjustment set. Hence $P(m \mid \mathrm{do}(T{=}t)) = P(m \mid T{=}t)$.
3. **Identify the effect of M on Y.** By condition 3, T is a valid back-door adjustment set for the $M \to Y$ effect:
$$P(y \mid \mathrm{do}(M{=}m)) \;=\; \sum_{t'} P(y \mid T{=}t',\; M{=}m)\, P(t').$$
4. **Substitute back** to obtain Equation 3.3. □

3.3.3 Interpretation: Two-Stage Causal Accounting

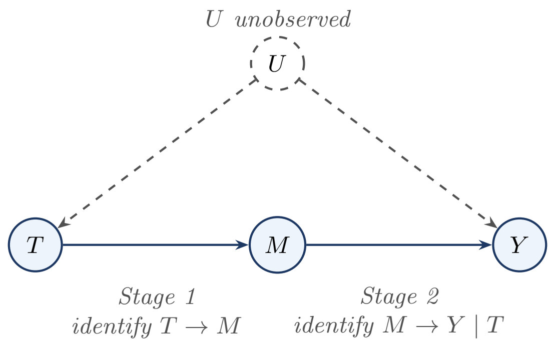


Figure 3.3: Why the front-door criterion works. The total effect of T on Y is not directly identifiable because U confounds T and Y. However, the effect of T on the mediator M is unconfounded (Stage 1), and the effect of M on Y becomes identifiable after conditioning on T, which blocks the back-door path through U (Stage 2).

Example: Smoking, Tar, and Cancer

The classic front-door example (Pearl 1995): T = smoking, M = tar deposits in lungs, Y = lung cancer, U = genetic predisposition (unobserved). The graph is $T \to M \to Y$ with $U \to T$ and $U \to Y$. There is no direct $T \to Y$ path.

Conditions 1–3 are satisfied: all causal paths from T to Y pass through M; no back-door path exists from T to M; and the only back-door path from M to Y is $M \leftarrow T \leftarrow U \to Y$, which is blocked by conditioning on T.

The front-door formula gives a double sum over (m, t') that is entirely computable from the observed joint distribution of (T, M, Y), without any data on U.

Example: Front-Door Formula — Binary Numerical Computation

Let $T, M, Y \in \{0, 1\}$ with U unobserved and the graph $T \to M \to Y$, $U \to T$, $U \to Y$.

T	M	Y	$P(T, M, Y)$
0	0	0	0.360
0	0	1	0.040
0	1	0	0.050

0	1	1	0.050
1	0	0	0.070
1	0	1	0.030
1	1	0	0.120
1	1	1	0.280

The inner sum (Stage 2):

$$S_0 = 0.1 \times 0.5 + 0.3 \times 0.5 = 0.20, \quad S_1 = 0.5 \times 0.5 + 0.7 \times 0.5 = 0.60.$$

The outer sum (Stage 1):

$$P(Y{=}1 \mid \mathrm{do}(T{=}1)) = 0.2 \times 0.20 + 0.8 \times 0.60 = 0.52,$$

$$P(Y{=}1 \mid \mathrm{do}(T{=}0)) = 0.8 \times 0.20 + 0.2 \times 0.60 = 0.28.$$

The average causal effect is $0.52 - 0.28 = 0.24$. The naïve estimate $P(Y{=}1 \mid T{=}1) - P(Y{=}1 \mid T{=}0) = 0.62 - 0.18 = 0.44$ is almost twice the true effect, because U simultaneously increases T and Y.

Comparing Back-Door and Front-Door

The two criteria are *not* nested. Neither implies the other.

- The back-door criterion requires all confounders to be observed (or blocked by observed variables), but places no restriction on mediators.
- The front-door criterion requires an observed mediator intercepting all causal paths, but the confounder U can be entirely unobserved.

There exist graphs where neither criterion alone applies, but where the full do-calculus (or the ID algorithm) can still identify the causal effect.

3.4 The Three Rules of Do-Calculus

Having seen two concrete identification strategies, we now develop the general algebraic machinery that underlies both. Pearl (1995) introduced the three rules of do-calculus as a system for manipulating expressions involving the do-operator: each rule licenses a transformation that removes or exchanges terms, bringing the expression one step closer to a purely observational formula.

How to Use a Do-Calculus Rule

When applying a do-calculus rule, proceed in three steps:

1. **Choose the relevant modified graph.** Form the graph obtained by deleting the appropriate incoming or outgoing arrows.
2. **Check the d-separation condition in that graph.** Work purely graphically: list the relevant paths and determine whether they are blocked.
3. **Apply the corresponding algebraic transformation.** Once the graphical condition holds, use the rule to remove an observation, insert an intervention, or exchange an observation for an intervention.

The do-calculus is therefore a graph modification $\to$ d-separation check $\to$ probability transformation procedure.

Remark: First Reading / Second Reading

On a first reading, focus on the qualitative role of the three rules: Rule 1 adds or removes observations, Rule 2 exchanges an action for an observation, and Rule 3 adds or removes actions. The detailed graph subscripts and formal statements can be absorbed gradually.

3.4.1 Statement of the Rules

Each rule's graphical condition uses a specific modified graph. The table below gives the recipe for each subscript that appears.

Subscript	**Step 1: delete**	**Step 2: delete**
$\mathcal{G}_{\overline{X}\underline{Z}}$	all arrows *into* X	all arrows *out of* Z
$\mathcal{G}_{\overline{X}\,\overline{Z(W)}}$	all arrows *into* X	arrows *into* Z except those from W
$\mathcal{G}_{\overline{X}\,\overline{Z(\overline{W})}}$	all arrows *into* X	arrows *into* Z that are not from ancestors of W in $\mathcal{G}_{\overline{X}}$

Theorem: The Three Rules of Do-Calculus

(Pearl 1995) Let $\mathcal{G}$ be a DAG over variables $\mathcal{V}$, and let $X, Y, Z, W \subseteq \mathcal{V}$ be disjoint sets. The rules hold for any distribution P Markov with respect to $\mathcal{G}$.
Rule 1 (Insertion/Deletion of Observations). *Remove or insert an observation when it becomes irrelevant in the modified graph.*

$$P(y \mid \mathrm{do}(x), z, w) \;=\; P(y \mid \mathrm{do}(x), w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\underline{Z}}}.$$

Rule 2 (Action/Observation Exchange). *Replace an intervention by an observation, or vice versa, when the modified graph makes them equivalent.*

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) \;=\; P(y \mid \mathrm{do}(x), z, w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\,\overline{Z(W)}}}.$$

Rule 3 (Insertion/Deletion of Actions). *Remove or insert an intervention when it becomes irrelevant in the modified graph.*

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) \;=\; P(y \mid \mathrm{do}(x), w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\,\overline{Z(\overline{W})}}}.$$

Remark

Each rule is a *conditional independence statement in a specific intervention graph.* The three rules cover all ways of adding or removing terms from a do-expression: Rule 1 handles observations; Rule 2 exchanges actions for observations; Rule 3 deletes actions. Any sequence of such operations that converts a do-expression into a purely observational expression is an *identification proof.*

Remark: Soundness of the Rules

The three rules are *sound*: each holds for every distribution P Markov with respect to $\mathcal{G}$. For Rule 2: the post-intervention distribution $P(\cdot \mid \mathrm{do}(x))$ is Markov with respect to $\mathcal{G}_{\overline{X}}$. Further intervening on Z gives a distribution Markov with respect to $\mathcal{G}_{\overline{X}\,\overline{Z}}$. The d-separation condition $(Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\,\overline{Z(W)}}}$ implies $Y \perp\!\!\!\perp Z \mid W$ in the post-do(x, z) distribution. Since the distribution of Y is the same whether we *intervene* on Z or *observe* Z, the two expressions are equal. Full proofs appear in Pearl (1995).

3.4.2 Intuition for Each Rule

Rule 1 — Adding/Removing Observations. The condition says: in the post-do(x) world, if we further deactivate Z's causal influence (delete its outgoing arrows), Y and Z become independent given W. This means Z carries no information about Y that is not already in W, so it can be freely inserted or removed.

Example: Rule 1 — Deleting a Redundant Observation

Graph: $Z \to T \to Y$, no other edges. Suppose we intervene on T.
In $\mathcal{G}_{\overline{T}\underline{Z}}$: delete arrows into T (removes $Z \to T$), then delete arrows out of Z (nothing remains). Node Z is isolated, so $(Y \perp\!\!\!\perp Z \mid T)$ holds trivially. Rule 1 gives:

$$P(y \mid \mathrm{do}(t), z) \;=\; P(y \mid \mathrm{do}(t)).$$

The intervention $\mathrm{do}(t)$ severs Z's only route to Y. Knowing the former cause Z of T is irrelevant once T has been set externally.

Rule 2 — Action/Observation Exchange. When the condition holds, intervening on Z is equivalent to conditioning on Z (within the scope of $\mathrm{do}(x)$ and W).

Example: Rule 2 — Exchanging Action for Observation

Graph: $X \to Z \to Y$, no other edges. Suppose we intervene on X and wish to show $P(y \mid \mathrm{do}(x), \mathrm{do}(z)) = P(y \mid \mathrm{do}(x), z)$.
In $\mathcal{G}_{\overline{X}\overline{Z}}$: delete arrows into X (removes the incoming arrows of X) and arrows into Z (removes $X \to Z$). Node Z has no incoming arrows, so no back-door paths from Z to Y can arise. The only path is the causal $Z \to Y$, and $(Y \perp\!\!\!\perp Z)_{\mathcal{G}_{\overline{X}\overline{Z}}}$ holds in the sense that no spurious association arises.
Rule 2 gives:

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z)) \;=\; P(y \mid \mathrm{do}(x), z).$$

Y depends on Z only through Z's value, not through the mechanism that generated Z. Externally fixing Z and merely observing $Z{=}z$ are therefore equivalent for Y.

Rule 3 — Deleting an Intervention. The condition says that Z has no causal path to Y that is not blocked by X or W in the appropriate intervention graph. Setting Z to any value has no effect on Y, so $\mathrm{do}(z)$ can be dropped.

Example: Rule 3 — Deleting an Irrelevant Intervention

Graph: $Z \to X \to Y$. Suppose we intervene on both X and Z.
In $\mathcal{G}_{\overline{X}\overline{Z}}$: delete arrows into X (removes $Z \to X$), then delete arrows into Z (none exist). Node Z is now isolated. Since there is no path from Z to Y, $(Y \perp\!\!\!\perp Z)_{\mathcal{G}_{\overline{X}\overline{Z}}}$ holds trivially. Rule 3 gives:

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z)) \;=\; P(y \mid \mathrm{do}(x)).$$

Once X is fixed externally, Z's only route to Y (through X) is severed. Setting Z to any value cannot change Y, so $\mathrm{do}(z)$ is superfluous.

Example: A Two-Rule Derivation — the Proxy Variable Graph

T has a direct effect on Y. An unobserved variable U is a common cause of T and an observed proxy S; S in turn causes Y. U does *not* directly cause Y. Edges: $T \to Y$, $U \to T$, $U \to S$, $S \to Y$.
Goal: simplify $P(y \mid \mathrm{do}(t))$ to a purely observational expression.
Step 1. Introduce S by the law of total probability:

$$P(y \mid \mathrm{do}(t)) = \sum_s P(y \mid \mathrm{do}(t),\, s)\; P(s \mid \mathrm{do}(t)).$$

Step 2. Rule 3 applied to $P(s \mid \mathrm{do}(t))$. Form $\mathcal{G}_{\overline{T}}$ by deleting the arrow $U \to T$. In this graph, T has no parents and no path connects T to S or to the U–S component. Hence $(S \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{T}}}$, and Rule 3 gives $P(s \mid \mathrm{do}(t)) = P(s)$.
Step 3. Back-door d-separation argument applied to $P(y \mid \mathrm{do}(t), s)$. Form $\mathcal{G}_{\underline{T}}$ by deleting the arrow $T \to Y$. The only back-door path $T \leftarrow U \to S \to Y$ is blocked by conditioning on S. Hence $\{S\}$ satisfies the back-door criterion, so $P(y \mid \mathrm{do}(t), s) = P(y \mid t, s)$.

Combining:

$$P(y \mid \mathrm{do}(t)) = \sum_s P(y \mid t,\, s)\; P(s).$$

The causal effect is identified by standardizing over the population marginal $P(s)$, even though the confounder U is entirely unobserved.

3.4.3 Proofs of the Identification Formulas

Proof: Back-Door Formula via Rule 3 and the Back-Door d-Separation Argument

Graph structure: $T \to Y$, unobserved $U \to T$, $U \to Y$, and observed $\mathbf{S}$ blocking the back-door path $T \leftarrow U \to Y$.
Step 1: Introduce S by the law of total probability.

$$P(y \mid \mathrm{do}(t)) = \int P(y \mid \mathrm{do}(t),\, \mathbf{s})\, dP(\mathbf{s} \mid \mathrm{do}(t)).$$

Step 2: Simplify $P(\mathbf{s} \mid \mathrm{do}(t))$. Because $\mathbf{S}$ contains no descendants of T (condition 1), the graphical condition for Rule 3 is $(\mathbf{S} \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{T}}}$: in $\mathcal{G}_{\overline{T}}$, all arrows into T are deleted so no directed path from T reaches any node in $\mathbf{S}$. Therefore $(\mathbf{S} \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{T}}}$ holds, and Rule 3 gives: $P(\mathbf{s} \mid \mathrm{do}(t)) = P(\mathbf{s})$.
Step 3: Convert $P(y \mid \mathrm{do}(t), \mathbf{s})$ to an observation. Condition 2 ensures that, within each stratum $\mathbf{S} = \mathbf{s}$, all back-door paths from T to Y are blocked. By the truncated factorization formula for $P(\cdot \mid \mathrm{do}(t))$, the post-intervention conditional equals the observational conditional: $P(y \mid \mathrm{do}(t), \mathbf{s}) = P(y \mid t, \mathbf{s})$.
Combining:

$$P(y \mid \mathrm{do}(t)) = \int P(y \mid t, \mathbf{s})\, dP(\mathbf{s}). \quad \square$$

Proof: Front-Door Formula via Three Applications of the Rules

Graph structure: $T \to M \to Y$, unobserved $U \to T$, $U \to Y$, no direct $T \to Y$ edge.
Step 1: Introduce M by the law of total probability.

$$P(y \mid \mathrm{do}(t)) = \int P(y \mid \mathrm{do}(t),\, m)\, dP(m \mid \mathrm{do}(t)).$$

Step 2: Simplify $P(m \mid \mathrm{do}(t))$. In $\mathcal{G}_{\underline{T}}$, there are no back-door paths from T to M by front-door condition 2. Rule 2 applies: $P(m \mid \mathrm{do}(t)) = P(m \mid t)$.
Step 3a: $P(y \mid \mathrm{do}(t), m) = P(y \mid \mathrm{do}(m))$. In $\mathcal{G}_{\overline{T}}$, the arrow $U \to T$ is deleted, so U and T become independent. Since there is no arrow from U into M (condition 2), the conditional of Y given $M{=}m$ integrates over U freely: $P(y \mid \mathrm{do}(t), m) = \int P(y \mid m, u)\, dP(u) = P(y \mid \mathrm{do}(m))$.
Step 3b: By front-door condition 3, $\{T\}$ satisfies the back-door criterion for the effect of M on Y, so $P(y \mid \mathrm{do}(m)) = \int P(y \mid m, t')\, dP(t')$.
Assembling:

$$P(y \mid \mathrm{do}(t)) = \int \left[\int P(y \mid t',\, m)\, dP(t') \right] dP(m \mid t). \quad \square$$

3.5 The Do-Calculus is Complete

A natural question after seeing the three rules is: are they enough? Could there be a graph where the causal effect is identifiable in principle, but none of the rules — applied in any sequence — can derive a purely observational expression? Shpitser and Pearl (2006) answered this definitively.

Throughout Sections 1–4 we have represented latent common causes as explicit unobserved nodes. The completeness theorem uses a more general graph class that encodes latent common causes compactly as *bidirected edges*: a bidirected edge $X \leftrightarrow Y$ stands for an unobserved variable U with $U \to X$ and $U \to Y$. A DAG augmented with such bidirected edges is called a *semi-Markovian DAG*.

Theorem: Completeness of the Do-Calculus

(Shpitser and Pearl 2006) Let $\mathcal{G}$ be a semi-Markovian DAG. A causal quantity $P(y \mid \mathrm{do}(t))$ is identifiable from P relative to $\mathcal{G}$ if and only if the do-calculus can derive a purely observational expression for it.

Example: The Bow Graph — The Simplest Non-Identifiable Structure

The *bow graph* is the semi-Markovian DAG with one directed edge $T \to Y$ and one bidirected edge $T \leftrightarrow Y$ (representing an unobserved U with $U \to T$ and $U \to Y$). The single back-door path is $T \leftarrow U \to Y$; no observed set can block it, so the back-door criterion cannot be satisfied. No observed mediator exists, so the front-door criterion also fails.
We show directly that $P(y \mid \mathrm{do}(t))$ is not determined by $P(T, Y)$ alone, by constructing two models that agree on $P(T, Y)$ but disagree on $P(y \mid \mathrm{do}(t))$.
Let $T, Y, U \in \{0, 1\}$ with $U \sim \text{Bernoulli}(1/2)$.
Model $\mathcal{M}_1$ (pure confounding; no causal effect of T): $T = U$, $Y = U$. Both T and Y are driven entirely by U. The observed joint distribution is $P(T{=}0, Y{=}0) = P(T{=}1, Y{=}1) = 1/2$ and $P(T \neq Y) = 0$. Under $\mathrm{do}(T{=}1)$, $Y = U$ is unchanged, so $P_1(Y{=}1 \mid \mathrm{do}(T{=}1)) = P(U{=}1) = 1/2$.
Model $\mathcal{M}_2$ (full causal effect; U acts on Y only through T): $T = U$, $Y = T$. Since $T = U$, we again have $Y = T = U$, so the observed joint distribution is *identical* to $\mathcal{M}_1$. Under $\mathrm{do}(T{=}1)$, $Y = T = 1$ deterministically, so $P_2(Y{=}1 \mid \mathrm{do}(T{=}1)) = 1$.
Conclusion. The two models produce identical observed distributions $P(T, Y)$, yet $P(Y{=}1 \mid \mathrm{do}(T{=}1))$ equals $1/2$ in $\mathcal{M}_1$ versus 1 in $\mathcal{M}_2$. Therefore $P(y \mid \mathrm{do}(t))$ is *not identified* from $P(T, Y)$ in the bow graph.

Proof Sketch and the ID Algorithm

The *ID algorithm* (Shpitser and Pearl 2006) takes as input a semi-Markovian DAG $\mathcal{G}$, a target intervention distribution $P(y \mid \mathrm{do}(t))$, and the observed joint distribution P, and either returns an explicit observational formula or reports non-identifiability.
Sufficiency (do-calculus identifies whenever identification is possible): proved constructively via the ID algorithm. The key structural concept is the *c-component* (confounded component): a maximal set of nodes connected by bidirected edges. The ID algorithm proceeds recursively: (1) simplification — remove from $\mathcal{G}$ any node that is not an ancestor of Y in $\mathcal{G}_{\overline{T}}$; (2) decomposition by c-components — factorise $P(\mathbf{v} \ \mathbf{t} \mid \mathrm{do}(t))$ over the c-components; (3) recursive identification — attempt to identify each c-component factor; (4) failure — if any subproblem cannot be reduced, the algorithm returns FAIL.
Necessity (do-calculus cannot identify non-identifiable quantities): proved by showing that whenever the ID algorithm fails, the causal effect is genuinely non-identifiable. The key graphical obstruction is the *hedge*: a pair of subgraphs (F, F') where F contains a back-door path from T to Y, and F' is a subgraph of $\mathcal{G}_{\overline{T}}$ in which Y remains connected to a latent common cause. The existence of a hedge allows explicit construction of two observationally equivalent models with different causal effects.
Combining: the do-calculus succeeds if and only if no hedge exists, which is if and only if $P(y \mid \mathrm{do}(t))$ is identifiable. □

Remark

The completeness theorem has a crucial practical implication: *if the ID algorithm reports non-identification, then no estimation method — however clever — can recover the causal effect from observational data alone*, given the assumed graph structure. Non-identification is not a limitation of a particular technique; it is a fundamental property of the causal model. Three remedies exist: impose additional structural assumptions (e.g., linearity or monotonicity) that go beyond the graph; collect additional data that break the obstruction (e.g., an instrument or a randomized experiment); or target a different, identified causal quantity instead.

3.6 The Big Picture

This chapter completes the *theoretical* identification machinery of the course. The chain of reasoning begun in Chapter 1 (the interventional/observational distinction) and formalized in Chapter 2 (d-separation) now terminates in a concrete identification formula. Chapters 4–7 then apply this machinery to specific research designs.

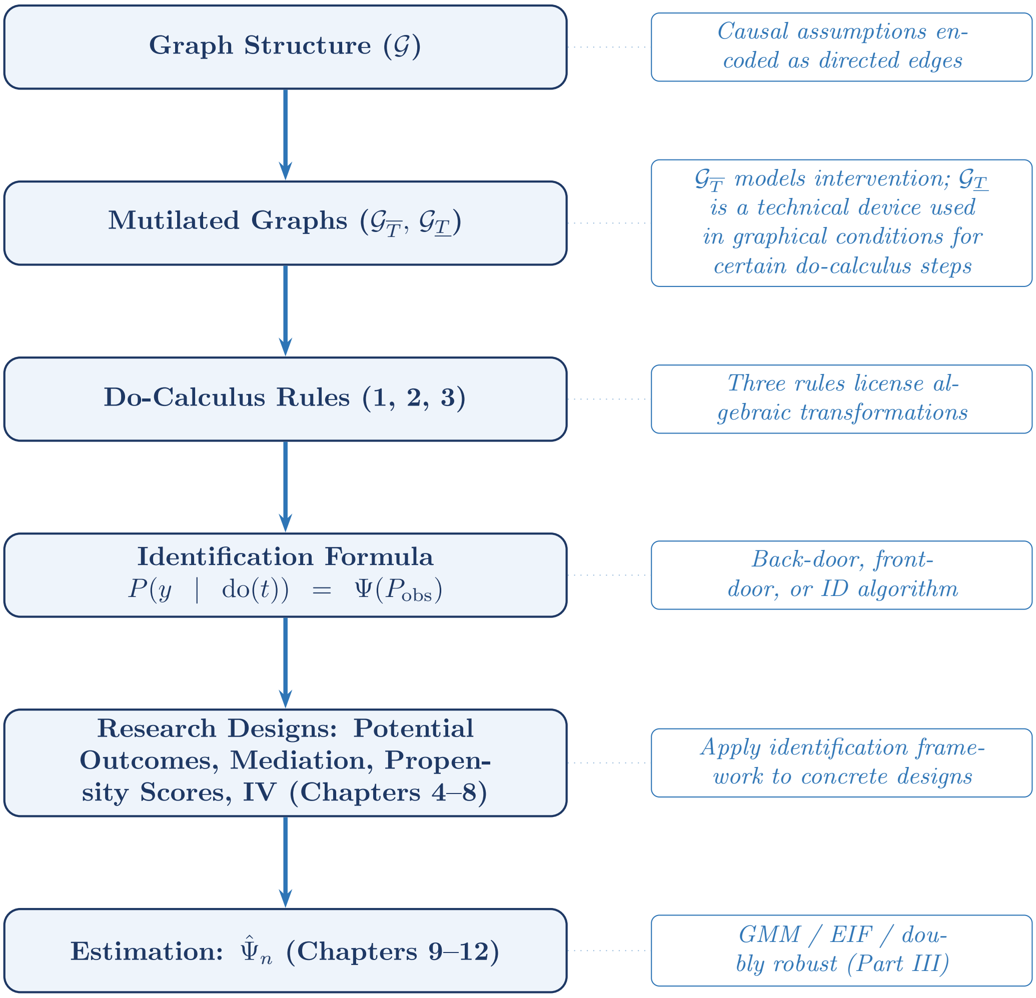


Figure 3.4: The six-step pipeline from causal assumptions to estimation.

The completeness theorem closes the loop: every causal effect that is identifiable from the observational distribution relative to the assumed graph can be derived by the do-calculus, and every effect that is not identifiable in that setting cannot be recovered by any observational method without additional assumptions or new data.

3.7 Summary

1. Two graph operations underpin the do-calculus: $\mathcal{G}_{\overline{X}}$ deletes arrows *into* X and models the intervention $\text{do}(x)$; $\mathcal{G}_{\underline{X}}$ deletes arrows *out of* X and is used as a technical device in graphical conditions for certain do-calculus steps — it does not model conditioning on X.
2. The back-door criterion identifies $P(y \mid \text{do}(t))$ when an observed set $\mathbf{S}$ blocks all back-door paths and contains no descendant of T. The general formula is $\int P(y \mid t, \mathbf{s})\, dP(\mathbf{s})$, the standardization formula (the g-formula for $K = 1$).
3. The front-door criterion identifies $P(y \mid \text{do}(t))$ via an observed mediator M intercepting all causal paths from T to Y, even when the confounder U is entirely unobserved, provided the three front-door conditions hold.

4. The do-calculus has three rules, each licensed by a d-separation condition in an appropriate intervention graph. Rule 1 adds/removes observations; Rule 2 exchanges actions for observations; Rule 3 deletes actions. The back-door and front-door formulas are each derivable as three-step sequences of these rules.

5. **Completeness** (Shpitser and Pearl 2006): the do-calculus is complete for identification from observational data relative to the assumed graph. A causal effect is identifiable if and only if the do-calculus can derive a purely observational expression for it. If the do-calculus cannot identify the quantity in that setting, then no purely observational method can do so without additional assumptions or new data.

3.8 Problems

1. Warm-up: graph operations. Let $\mathcal{G}$ be the IV DAG with edges $Z \to T$, $T \to Y$, $U \to T$, $U \to Y$ (U unobserved).

(a) Draw $\mathcal{G}_{\overline{T}}$ and identify all paths from Z to Y that remain open.
(b) Draw $\mathcal{G}_{\underline{T}}$ and identify all paths from Z to Y that remain. For each path, classify it as a causal chain, fork, or collider path, and state whether it is open or closed by default.
(c) In $\mathcal{G}_{\overline{T}}$, is $Z \perp\!\!\!\perp Y \mid \emptyset$? How does the absence of a direct edge $Z \to Y$ in the DAG relate to the exclusion restriction?

2. Back-door practice. Consider the DAG: $X \to T$, $X \to Y$, $T \to M$, $M \to Y$, $T \to Y$, where X is observed.

(a) List all back-door paths from T to Y.
(b) Does $\{X\}$ satisfy the back-door criterion for the effect of T on Y? Write the resulting adjustment formula.
(c) Does $\{M\}$ satisfy the back-door criterion? Explain why or why not.
(d) Does $\{X, M\}$ satisfy the back-door criterion? Identify which condition of the criterion M violates.
(e) Even if the formal criterion issue in (d) were somehow set aside, explain the substantive bias that arises from conditioning on a mediator M that lies on a causal path $T \to M \to Y$.

3. Front-door identification. Consider the DAG: $T \to M \to Y$, with $U \to T$ and $U \to Y$ (unobserved U, no direct $T \to Y$ edge).

(a) Verify that M satisfies all three front-door conditions.
(b) Derive the front-door formula Equation 3.3 step by step, citing the do-calculus rule used at each step.
(c) Now add a direct edge $T \to Y$ to the graph. Does M still satisfy the front-door criterion? Explain which condition fails.

4. Identification or non-identification? For each of the following DAGs, determine whether $P(y \mid \mathrm{do}(t))$ is identified non-parametrically from observational data. If identified, state the formula; if not, explain the structural obstruction.

(a) $T \to Y$, $U \to T$, $U \to Y$, U unobserved.
(b) Same as (a), with instrument $Z \to T$ and $Z \perp\!\!\!\perp U$ (no direct $Z \to Y$ edge). Show that $P(y \mid \mathrm{do}(t))$ is **not** identified non-parametrically from $P(Z, T, Y)$ by constructing two binary SEMs $\mathcal{M}_1$ and $\mathcal{M}_2$, each compatible with the IV graph, such that (i) $P_{\mathcal{M}_1}(Z, T, Y) = P_{\mathcal{M}_2}(Z, T, Y)$ and (ii) $P_{\mathcal{M}_1}(Y{=}1 \mid \mathrm{do}(T{=}1)) \neq P_{\mathcal{M}_2}(Y{=}1 \mid \mathrm{do}(T{=}1))$. (*Hint*: let $Z, U \sim \mathrm{Bern}(1/2)$ independently and set $T = U$ in both models. Then set $Y = U$ in $\mathcal{M}_1$ and $Y = T$ in $\mathcal{M}_2$.)
(c) $T \to M \to Y$, $U \to T$, $U \to Y$, $U \to M$, U unobserved.
(d) $T \to Y$, $X \to T$, $X \to Y$, X observed.

5. Back-door formula: proof and uniqueness.

(a) Let $\mathcal{M}$ be a structural causal model Markov with respect to $\mathcal{G}$, and let $\mathbf{S}$ satisfy the back-door criterion for (T, Y) in $\mathcal{G}$ with positivity. Prove that $P(y \mid \mathrm{do}(t)) = \int P(y \mid t, \mathbf{s})\, dP(\mathbf{s})$. (*Hint*: begin with $P(y \mid \mathrm{do}(t)) = \int P(y \mid \mathrm{do}(t), \mathbf{s})\, P(\mathbf{s} \mid \mathrm{do}(t))\, d\mathbf{s}$. Apply the Rule 3 graphical condition $(\mathbf{S} \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{T}}}$, using back-door condition 1, to simplify $P(\mathbf{s} \mid \mathrm{do}(t))$. Then invoke the d-separation $(Y \perp\!\!\!\perp T \mid \mathbf{S})_{\mathcal{G}_{\underline{T}}}$, which back-door condition 2 implies, together with the truncated factorization, to argue $P(y \mid \mathrm{do}(t), \mathbf{s}) = P(y \mid t, \mathbf{s})$.)

(b) Suppose $\mathbf{S}_1$ and $\mathbf{S}_2$ both satisfy the back-door criterion for (T, Y) in $\mathcal{G}$ (with positivity). Deduce from part (a) that $\int P(y \mid t, \mathbf{s}_1)\, dP(\mathbf{s}_1) = \int P(y \mid t, \mathbf{s}_2)\, dP(\mathbf{s}_2)$. Interpret this result: the identification target $P(y \mid \mathrm{do}(t))$ is unique even when the adjustment set is not.

6. Rule sequencing on an unseen graph. Consider the graph: $C_1 \to T$, $C_1 \to Y$, $U \to T$, $U \to C_2$, $C_2 \to Y$, $T \to Y$, with C_1 and C_2 observed and U unobserved.

Derive $P(y \mid \mathrm{do}(t)) = \sum_{c_1, c_2} P(y \mid t, c_1, c_2)\, P(c_1, c_2)$ from first principles using the three rules. For each step: (i) name the rule applied, (ii) form the relevant intervention graph by stating which arrows are deleted and what edges remain, and (iii) verify the required d-separation condition by tracing all paths and confirming each is blocked.

7. Non-identifiability by construction. Consider the graph from the front-door failure example: $T \to M \to Y$, $U \to T$, $U \to M$, $U \to Y$ (front-door condition 2 violated). Let $T, M, Y, U \in \{0, 1\}$ with $U \sim \mathrm{Bern}(1/2)$.

(a) Construct two SEMs $\mathcal{M}_1$ and $\mathcal{M}_2$, each compatible with the graph, such that $P_{\mathcal{M}_1}(T, M, Y) = P_{\mathcal{M}_2}(T, M, Y)$ but $P_{\mathcal{M}_1}(Y{=}1 \mid \mathrm{do}(T{=}1)) \neq P_{\mathcal{M}_2}(Y{=}1 \mid \mathrm{do}(T{=}1))$. (*Hint*: set $T = U$ in both models. In $\mathcal{M}_1$ set $M = U$ and $Y = U$; in $\mathcal{M}_2$ set $M = T$ and $Y = M$.)

(b) Explain precisely which identification strategy fails in this graph and why, referencing (i) the back-door criterion, (ii) the front-door criterion (which condition fails and which stage collapses), and (iii) the completeness theorem (what does the existence of your two models imply?).

Chapter 4

Potential Outcomes and the Connection to Do-Calculus

Learning Objectives

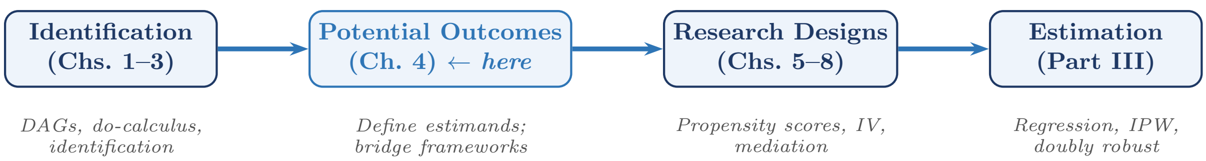
By the end of this chapter, students should be able to:

1. Define the potential outcome $Y(t)$, state SUTVA precisely, and derive the consistency equation $Y = Y(T)$.
2. Define the ATE and ATT, explain how they relate to each other, and re-express them as functionals of the structural equation for Y.
3. Show that the linear SEM structural coefficient β equals the ATE under homogeneous effects, and explain why ATE and ATT diverge under treatment effect heterogeneity.
4. State the ignorability assumption, interpret it graphically as a back-door condition, and explain why overlap is a necessary companion assumption.
5. Describe the SWIG construction of Richardson and Robins (2014), explain why the fixed half is a source node, and use d-separation in the SWIG to derive the ignorability condition directly from the graph.

4.1 Motivation: A Third Language for Causality

The first three chapters developed causal inference primarily in the languages of structural equations and directed acyclic graphs. In this chapter we introduce a third language: the potential outcomes framework of Neyman (1923) and Rubin (1974).

Potential outcomes provide a direct language for defining causal estimands. Quantities such as the average treatment effect, the average treatment effect on the treated, and related causal contrasts are most naturally written in terms of counterfactual outcomes $Y(1)$, $Y(0)$, and more generally $Y(t)$. The two frameworks should be viewed as complementary rather than competing: potential outcomes define the causal quantities of interest, while DAGs and the do-calculus clarify identification.



4.2 The Neyman–Rubin Potential Outcomes Framework

4.2.1 The Potential Outcome

Definition: Potential Outcome

(Neyman 1923; Rubin 1974) For each unit i and each possible treatment value $t \in \mathcal{T}$, the *potential outcome* $Y_i(t)$ is the value of the outcome that unit i *would have exhibited* had its treatment been set to t, possibly contrary to fact.
The collection $\{Y_i(t) : t \in \mathcal{T}\}$ is called the *schedule of potential outcomes* for unit i. In the binary case $\mathcal{T} = \{0, 1\}$, the schedule is $(Y_i(0), Y_i(1))$.

$Y_i(1)$ is the outcome unit i would achieve *under treatment*; $Y_i(0)$ is the outcome under *control*. Only one of these is ever observed for any given unit. The unobserved potential outcome is called the *counterfactual*.

The Fundamental Problem of Causal Inference

(Holland 1986) For any unit i, at most one potential outcome $Y_i(t)$ is observed. The individual-level causal effect $Y_i(1) - Y_i(0)$ is therefore *never directly observable*. Causal inference is an inference problem precisely because this quantity must be recovered from a population of units rather than from a single unit.

4.2.2 SUTVA and Consistency

Definition: SUTVA

(Rubin 1980) The *stable unit treatment value assumption* has two components:

1. **No interference.** The potential outcome $Y_i(t)$ depends only on unit i's own treatment, not on the treatments of other units: $Y_i(t_1, \dots, t_n) = Y_i(t_i)$.
2. **No hidden versions.** For each treatment level t, there is a single well-defined version.

The consistency equation. Under SUTVA, the observed outcome Y_i equals the potential outcome at the treatment actually received:

$$Y_i \;=\; Y_i(T_i). \tag{4.1}$$

This equation is the bridge between the potential outcome world and the observed data world. It fails whenever SUTVA fails: if treatment spills over between units (e.g., vaccination provides herd immunity), or if the treatment label $T = 1$ covers multiple distinct interventions.

4.2.3 Connection to the Do-Operator

The notation $Y(t)$ and the expression $P(Y \mid \mathrm{do}(T{=}t))$ refer to the same causal idea viewed from two complementary angles. Under the structural causal framework adopted in these notes, together with consistency and no interference, these two descriptions coincide.

Proposition: Potential Outcomes and the Do-Operator

Under the structural causal model of Chapters 1–3, together with consistency and no interference (SUTVA),

$$Y(t) \;\stackrel{d}{=}\; Y \mid \mathrm{do}(T{=}t), \tag{4.2}$$

so that $P(Y(t) \le y) = P(Y \le y \mid \mathrm{do}(T{=}t))$ for every y. Equivalently, $\mathbb{E}[Y(t)] = \mathbb{E}[Y \mid \mathrm{do}(T{=}t)]$.

Proof

In the mutilated SEM $\mathcal{G}_{\overline{T}}$, the structural equation for T is replaced by $T := t$ for every unit. The structural model then determines the value of Y for unit i from t and unit i's own background variables — which is precisely what the potential outcomes framework defines as $Y_i(t)$. SUTVA's no-interference condition ensures $Y_i(t)$ does not depend on other units' treatment values, so the marginal distribution of Y in $\mathcal{G}_{\overline{T}}$ equals the marginal distribution of $Y(t)$. By definition of the do-operator, the former is $f(y \mid \mathrm{do}(T{=}t))$. $\square$

This equivalence lets us move freely between two notational traditions. When defining estimands such as $\mathbb{E}[Y(1)-Y(0)]$, potential-outcome notation is often most natural. When proving identification results from a graph, the do-operator is often more transparent: $f(y \mid \mathrm{do}(t)) \neq f(y \mid T{=}t)$ whenever T is endogenous, whereas $Y(t)$ carries no such syntactic marker.

4.3 Causal Estimands

4.3.1 The Average Treatment Effect and Its Relatives

Definition: ATE and ATT

For a binary treatment $T \in \{0,1\}$:

- The *average treatment effect* (ATE) is $\tau_{\mathrm{ATE}} = \mathbb{E}[Y(1) - Y(0)]$.
- The *average treatment effect on the treated* (ATT) is $\tau_{\mathrm{ATT}} = \mathbb{E}[Y(1) - Y(0) \mid T{=}1]$.

The ATE and ATT coincide only when the treatment effect does not depend on who selected into treatment. The naïve estimator $\hat{\tau}_{\mathrm{naive}} = \bar{Y}_{T=1} - \bar{Y}_{T=0}$ estimates $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0] \neq \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$, with the gap being the selection bias induced by endogeneity.

Remark

ATE and ATT coincide when $\mathbb{E}[Y(t) \mid T] = \mathbb{E}[Y(t)]$ for $t \in \{0,1\}$, i.e., when treatment assignment is mean-independent of potential outcomes. In a randomized experiment, this holds by design.

4.3.2 Causal Estimands as Functionals of the Structural Equation

The potential outcome $Y_i(t)$ was introduced as a *hypothetical*: the value unit i would have exhibited under treatment t. The SEM framework of Chapter 1 gives this hypothetical a precise generative meaning.

From structural equation to potential outcome. In the SEM, the outcome is determined by a structural equation

$$Y \;=\; g(T,\, \mathbf{X},\, U_Y), \tag{4.3}$$

where U_Y collects all sources of variation in Y not already accounted for by $(T, \mathbf{X})$. The potential outcome under $\mathrm{do}(T{=}t)$ is obtained by substituting t for T while holding everything else fixed:

$$Y_i(t) \;=\; g(t,\, \mathbf{X}_i,\, U_{Y,i}). \tag{4.4}$$

ATE and ATT as structural parameters.

$$\tau_{\mathrm{ATE}} = \mathbb{E}\left[g(1, \mathbf{X}, U_Y) - g(0, \mathbf{X}, U_Y)\right], \tag{4.5}$$

$$\tau_{\mathrm{ATT}} = \mathbb{E}\left[g(1, \mathbf{X}, U_Y) - g(0, \mathbf{X}, U_Y) \mid T{=}1\right]. \tag{4.6}$$

The linear SEM as a special case. In the Gaussian linear SEM of Chapter 1, $Y = \beta T + \gamma^\top \mathbf{X} + \varepsilon$, the unit-level effect is $Y_i(1) - Y_i(0) = \beta$ for every unit. Consequently,

$$\tau_{\mathrm{ATE}} \;=\; \tau_{\mathrm{ATT}} \;=\; \beta.$$

The structural coefficient β *is* the average treatment effect: no averaging over heterogeneity is needed because there is none.

Heterogeneous effects. In the nonparametric SEM Equation 4.3, the unit-level effect $g(1, \mathbf{X}_i, U_{Y,i}) - g(0, \mathbf{X}_i, U_{Y,i})$ varies across units. The ATE averages this over the full population; the ATT averages it over the treated subpopulation. The two differ whenever treatment selection correlates with the individual effect size.

Remark

The SEM representation Equation 4.4 clarifies why the potential outcomes framework and the do-calculus are *equivalent in content but different in emphasis*. The do-calculus works with the interventional distribution $f(y \mid \mathrm{do}(T{=}t))$ as a population-level object. The potential outcomes framework works with unit-level quantities $Y_i(t)$ and asks what population summaries (ATE, ATT) are scientifically meaningful. The SEM ties the two together: $Y_i(t) = g(t, \mathbf{X}_i, U_{Y,i})$ is the unit-level object, and integrating over the distribution of $(\mathbf{X}, U_Y)$ recovers the interventional distribution.

4.4 Ignorability and the Back-Door Criterion

4.4.1 The Ignorability Assumption

The central identifying assumption in observational studies is that treatment assignment is as good as random after conditioning on observed covariates X.

Definition: Strong Ignorability

(Rosenbaum and Rubin 1983) The treatment assignment T is *strongly ignorable* given X if:

1. **Unconfoundedness:** $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$.
2. **Overlap (positivity):** $0 < P(T{=}1 \mid X{=}x) < 1$ for all x in the support of X.

Under strong ignorability, the ATE is identified:

$$\tau_{\mathrm{ATE}} \;=\; \mathbb{E}_X[\mathbb{E}[Y \mid T{=}1, X] - \mathbb{E}[Y \mid T{=}0, X]]\,. \tag{4.7}$$

This is the back-door adjustment formula of Chapter 3, written in potential-outcome notation.

4.4.2 Faithfulness and the Completeness of D-Separation

Definition: Faithfulness

A distribution P is **faithful** to a DAG $\mathcal{G}$ if every conditional independence that holds in P is entailed by d-separation in $\mathcal{G}$: that is, $X \perp\!\!\!\perp Y \mid \mathbf{Z}$ in P only if X and Y are d-separated by $\mathbf{Z}$ in $\mathcal{G}$. Equivalently, faithfulness rules out *accidental* cancellations of causal paths. Together, the Markov property and faithfulness make d-separation a *complete* characterization of the conditional independence structure of P.

Faithfulness is deferred to this chapter because its primary role is as a hypothesis of the theorem below, not as a tool for reading conditional independences off a graph. In practice, faithfulness fails only on measure-zero sets of parameter values, so it is considered a mild assumption.

4.4.3 Ignorability as a Graphical Statement

The potential-outcomes assumption $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ is the counterfactual expression of the idea that, after conditioning on X, treatment assignment carries no residual information about the outcome that would be observed under the intervention $T{=}t$. In the language of DAGs, the closely related idea is that X blocks all back-door paths from T to Y.

Theorem: Ignorability $\Leftrightarrow$ Back-Door Criterion

Under the Markov and faithfulness assumptions for the DAG $\mathcal{G}$, unconfoundedness $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ is equivalent to X satisfying the back-door criterion for the effect of T on Y in $\mathcal{G}$:

1. No node in X is a descendant of T.
2. X blocks every back-door path from T to Y.

Proof Sketch

($\Rightarrow$) Suppose X satisfies the back-door criterion. In the mutilated graph $\mathcal{G}_{\overline{T}}$, all arrows into T are deleted, severing every back-door path from T to Y. The back-door criterion then guarantees that X blocks all remaining non-causal paths between T and Y in $\mathcal{G}_{\overline{T}}$. The Markov property applied to $\mathcal{G}_{\overline{T}}$ gives $(Y \perp\!\!\!\perp T \mid X)_{\mathcal{G}_{\overline{T}}}$, which translates to $(Y(t) \perp\!\!\!\perp T \mid X)$ for all t, i.e., unconfoundedness.
($\Leftarrow$) Suppose $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ holds. By faithfulness, every conditional independence in P corresponds to a d-separation in $\mathcal{G}$. Consider the SWIG $\mathcal{G}(t)$ obtained by splitting T into its random and fixed halves. Because the fixed half t has no incoming edges by construction, every path from the random half T to a potential outcome $Y(t)$ must pass through T's parents in $\mathcal{G}$ — that is, every such path is a back-door path. The d-separation $(Y(t) \perp\!\!\!\perp T \mid X)_{\mathcal{G}(t)}$ therefore requires X to block all back-door paths from T to Y. If any member of X were a descendant of T, conditioning on it could open a previously blocked collider path, contradicting the assumed independence. Both conditions of the back-door criterion are satisfied. $\square$

Remark

The direction most important for practice is ($\Rightarrow$): a graphical adjustment set justifies the counterfactual independence needed for standardization, regression adjustment, or propensity-score methods, and it requires only the Markov property. In applied work it is safest to regard the back-door criterion as a *sufficient* graphical condition for ignorability, and to invoke the biconditional only when faithfulness has been explicitly assumed.

Example: Labor Training Program

(LaLonde 1986; Dehejia and Wahba 1999) Let $T =$ receipt of job training, $Y =$ earnings two years later, $X =$ (age, education, prior earnings, race, marital status).
The back-door path $T \leftarrow X \rightarrow Y$ is blocked by conditioning on X, leaving only the causal path $T \rightarrow Y$. Ignorability holds if this DAG is correctly specified. If there is an unobserved variable U (motivation, ability) that affects both training participation and earnings, X fails the back-door criterion and ignorability fails.

4.4.4 Overlap and Positivity

Why Overlap Is Non-Negotiable

Without overlap, some covariate strata contain only treated or only control units. In those strata, $\mathbb{E}[Y \mid T{=}0, X{=}x]$ or $\mathbb{E}[Y \mid T{=}1, X{=}x]$ is unobservable, and Equation 4.7 cannot be evaluated at those values of x. Identification of the ATE fails not because the causal structure is wrong, but because the data do not span the support needed. The ATT may remain identified even when full overlap fails, because ATT averages only over the treated population where $P(T{=}1 \mid X) > 0$ by construction.

4.5 Single World Intervention Graphs

The potential outcomes framework and the do-calculus share a common goal but differ in representation. Single World Intervention Graphs (SWIGs), introduced by Richardson and Robins (2014) and given an accessible practical treatment by Bezuidenhout et al. (2025), provide a unified graphical representation that encodes both.

4.5.1 The SWIG Construction

In the original DAG $\mathcal{G}$, the node T represents the treatment as it naturally occurs. Under do($T{=}t$), however, the treatment is externally fixed. The SWIG separates these two roles by *splitting* T into two pieces:

- a **random half** (left side, capital letter T): represents the treatment value that would naturally occur, and retains all incoming edges from T's causal parents;
- a **fixed half** (right side, lowercase letter t): represents the externally imposed intervention value, carries all outgoing edges that formerly left T, and has *no* edge to or from the random half.

Definition: Single World Intervention Graph

(Richardson and Robins 2014) The *SWIG* $\mathcal{G}(t)$ is constructed from $\mathcal{G}$ by:

1. Replacing T with two nodes: random half T (all incoming edges retained) and fixed half t (all outgoing edges retained).
2. Severing the direct connection between the two halves.
3. Redrawing causal arrows at the split node: *all arrows into the split node land on the random half*; *all arrows departing the split node leave from the fixed half.*
4. Relabeling every descendant V of T in $\mathcal{G}$ as $V(t)$, indicating it is a potential outcome under do(T=t).

Figure 4.1 illustrates the construction for the simple confounded-treatment graph.

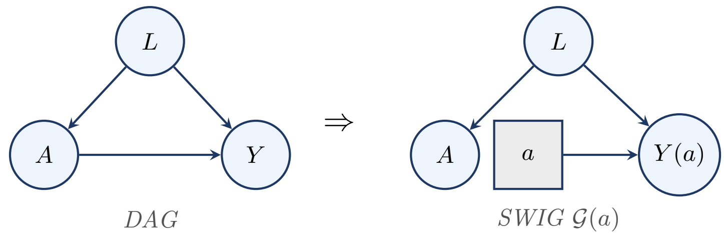


Figure 4.1: Converting the confounded-treatment DAG (L is a confounder) into the SWIG for do(A=a). The treatment node splits: the random half A (circle) retains the incoming arrow from L; the fixed half a (square) carries the outgoing arrow to $Y(a)$. D-separation in the SWIG gives $A \perp\!\!\!\perp Y(a) \mid L$ (conditional ignorability).

4.5.2 The Fixed Half as a Source Node

The Fixed Half Is a Source Node

In $\mathcal{G}(t)$, the fixed half t has **no incoming edges**. Step 2 of the construction severs the connection from the random half T, and no other node in $\mathcal{G}$ has a direct arrow into t. Three consequences follow immediately.

1. **No path arrives at** t **from another node.** The edge $T \to t$ does not exist in $\mathcal{G}(t)$. In particular, the path $T \to t \to Y(t)$ *cannot be formed*: it is not "blocked" by a d-separation rule — it simply does not exist as a path in the graph.
2. t **cannot lie strictly between two other nodes on a path.** Because t has no parents, it is never a transit node; it can only be an *origin* of paths leading to descendants $V(t)$.
3. **The random half** T **and the fixed half** t **are d-separated by the empty set.** With no edge between them and no common ancestor, $T \perp\!\!\!\perp t$ holds unconditionally in $\mathcal{G}(t)$.

This is not an additional d-separation rule: it is a direct consequence of the SWIG construction. The fixed half t represents an externally stipulated intervention — it is not *caused* by anything within the system, so it has no parents.

With t as a source node, every path from the random half T to a potential outcome $V(t)$ must travel *away* from T through T's parents in $\mathcal{G}$ — that is, every such path is a back-door path. This is the structural fact that makes the back-door criterion readable directly from the SWIG.

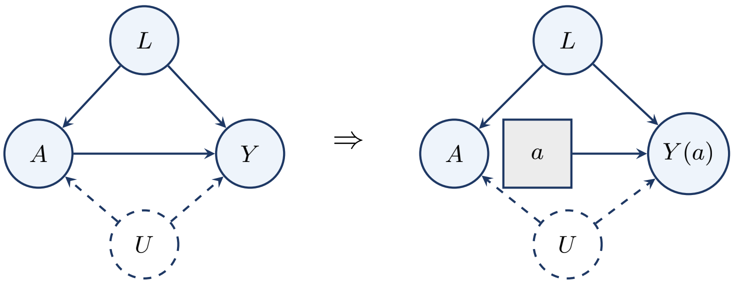
Recovering ignorability from the SWIG. In Figure 4.1, the only paths from the random half A to $Y(a)$ are back-door paths through A's parents. The path $A \leftarrow L \to Y(a)$ is open, but blocked by conditioning on L. No other path connects A to $Y(a)$. Conditioning on L achieves d-separation:

$$A \perp\!\!\!\perp Y(a) \mid L \quad \text{in } \mathcal{G}(a),$$

which is exactly the conditional ignorability of Definition 4.3 (Strong Ignorability), now *derived* as a graph-structural consequence rather than asserted as an assumption.

4.5.3 When Ignorability Fails: The Hidden Confounder

The SWIG is equally useful for diagnosing *failures* of ignorability. Figure 4.2 shows the SWIG when a hidden common cause U affects both A and Y.



DAG (with hidden U) *SWIG $\mathcal{G}(a)$ (with hidden U)*

Figure 4.2: Adding a hidden confounder U (dashed circle) to the confounded-treatment DAG and its SWIG. In $\mathcal{G}(a)$, the hidden path $A \leftarrow U \rightarrow Y(a)$ cannot be blocked by any observed covariate. Ignorability fails.

In the SWIG $\mathcal{G}(a)$ with hidden U, the paths from A to $Y(a)$ are: (1) $A \leftarrow L \rightarrow Y(a)$: blocked by conditioning on L; (2) $A \leftarrow U \rightarrow Y(a)$: open regardless of L, since U is unobserved.

Even after conditioning on L, the path through U remains open, so $A \not\perp\!\!\!\perp Y(a) \mid L$ in $\mathcal{G}(a)$, and ignorability fails. The failure is *immediately visible* from the SWIG: an unblocked dashed arrow entering the random half A signals that selection into treatment carries information about the potential outcome that no set of observed covariates can remove.

> Remark
>
> This failure motivates the instrumental variable designs of Chapter 7. When a valid instrument Z is available — one that affects A but has no direct effect on Y and is independent of U — the IV strategy sidesteps the unblocked U-path rather than attempting to block it by covariate adjustment.

4.5.4 What SWIGs Achieve

1. Potential outcomes as random variables in a DAG. In $\mathcal{G}(t)$, the variable $Y(t)$ is an ordinary node with well-defined parents: the fixed half t and any other causes of Y retained from $\mathcal{G}$. Its marginal distribution equals the interventional distribution $f(y \mid \mathrm{do}(T{=}t))$, establishing Proposition 4.1 graphically.

2. The mutilated graph as a special case. Removing the random half T from $\mathcal{G}(t)$ yields exactly the mutilated graph $\mathcal{G}_{\overline{T}}$ of Chapter 1. The SWIG is a more informative version that retains the random half alongside the fixed half, making it possible to ask questions about the relationship between the natural treatment T and the potential outcomes $V(t)$.

3. Unconfoundedness as a d-separation statement. The ignorability condition $(Y(t) \perp\!\!\!\perp T \mid X)$ is a standard d-separation statement *in the SWIG* $\mathcal{G}(t)$, verifiable by path-tracing. Ignorability is no longer an assertion encoded in potential-outcome notation alone; it is a structural claim about the graph.

4. A single graph for both frameworks. The SWIG simultaneously hosts the natural treatment T (random half), the intervention value t (fixed half), the potential outcomes $V(t)$, and all d-separation relationships among them.

Cross-World Independence and Its Limits

Consider $Y(1) \perp\!\!\!\perp Y(0)$: $Y(1)$ lives in SWIG $\mathcal{G}(1)$ and $Y(0)$ lives in SWIG $\mathcal{G}(0)$. No unit is ever observed in both worlds simultaneously, so no data can directly test this independence. More generally, statements such as $Y(t) \perp\!\!\!\perp T(t') \mid X$ for $t \neq t'$ involve variables from *different SWIGs*. SWIGs make such assumptions explicit: the analyst must specify a joint distribution across multiple SWIGs — a commitment that potential-outcome notation alone does not force.

Remark

More than one node may be split in a single SWIG. In longitudinal settings with time-varying treatments (A_0, A_1), splitting both nodes yields the sequential-randomization SWIG, from which the two *sequential exchangeability* conditions needed to identify the g-formula follow directly by d-separation.

4.6 Where the Frameworks Agree and Diverge

Task	**Potential Outcomes**	**Do-Calculus / DAG**	**SEM**
Define causal estimands (ATE, ATT)	Native language	Via $\mathbb{E}[Y \mid \mathrm{do}(t)]$	Via structural equations
Encode causal assumptions	Stated informally; graph hidden	Explicit directed edges	Structural equations
Read conditional independence	Not native without additional structure	d-separation	With graph only
Identification from observational data	Via ignorability assumptions	Back-door, front-door, do-calculus	Via exclusion restrictions
Cross-world assumptions (e.g., monotonicity)	Natural to state	Requires multiple SWIGs	Implicit in structural model
Standard in statistics/epidemiology	Dominant	Growing rapidly	Econometrics

Our Position in This Course

Both frameworks are indispensable. We use *potential outcomes* to *define* causal estimands. We use the *do-calculus and DAGs* for *identification*. SWIGs provide the formal bridge when both are needed simultaneously.
The do-operator language is preferred throughout this course because it makes the interventional/observational distinction *syntactically enforced*: $f(y \mid \mathrm{do}(t))$ cannot be confused with $f(y \mid T{=}t)$, whereas $\mathbb{E}[Y(t)]$ carries no such syntactic marker.

4.7 Summary

1. The potential outcome $Y(t)$ is the outcome that would be observed under the intervention $T = t$. Under consistency, no interference, and the structural causal semantics adopted in these notes, the observed outcome satisfies $Y_i = Y_i(T_i)$, and the potential outcome $Y(t)$ has the same distribution as the outcome under the intervention $\mathrm{do}(T{=}t)$.

2. The ATE and ATT are averages of the unit-level causal effect $Y_i(1) - Y_i(0) = g(1, \mathbf{X}_i, U_{Y,i}) - g(0, \mathbf{X}_i, U_{Y,i})$ over the full population and the treated subpopulation respectively. In the linear SEM, both equal the structural coefficient β. They diverge under heterogeneous effects when treatment selection correlates with individual effect size.

3. Strong ignorability $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ with overlap identifies the ATE via Equation 4.7. The back-door criterion provides a sufficient graphical condition for the conditional exchangeability assumption used in adjustment formulas.

4. SWIGs (Richardson and Robins 2014; Bezuidenhout et al. 2025) are constructed from a DAG by splitting each intervened node into a random half (capital letter, retaining incoming edges) and a fixed half (lowercase letter, retaining outgoing edges), and relabeling descendants as potential outcomes. The fixed half is a *source node*: it has no incoming edges by construction, so no path from the random half T to a potential outcome $V(t)$ can pass through t. Every such path is therefore a back-door path, and d-separation in the SWIG makes ignorability graphically transparent.

5. The frameworks are complementary: potential outcomes define estimands; the do-calculus identifies them. SWIGs provide the formal bridge when both are needed simultaneously.

In observational studies, ignorability must be justified by substantive knowledge encoded in a causal graph. Randomized experiments provide a fundamentally different solution: when treatment is assigned randomly, $(Y(0), Y(1)) \perp\!\!\!\perp T$ holds by design, guaranteeing ignorability without any covariate adjustment. Chapter 5 studies randomized experiments as the canonical design where causal effects are identifiable directly from the data-generating mechanism.

4.8 Problems

1. SUTVA and consistency. Suppose $n = 3$ units receive binary treatments (T_1, T_2, T_3) and unit i's outcome may depend on all three treatments: $Y_i = Y_i(T_1, T_2, T_3)$.

(a) How many potential outcomes does unit 1 have? Write them out explicitly.
(b) SUTVA imposes no interference: $Y_i(T_1, T_2, T_3) = Y_i(T_i)$. How many distinct potential outcomes remain?
(c) Give a real-world example where no-interference plausibly holds and one where it plausibly fails. In the latter case, explain what identification strategy (if any) remains available.

2. ATE, ATT, and selection bias. Let $(Y(0), Y(1), T) \sim P$ with $P(T{=}1) = 0.5$. Suppose $\mathbb{E}[Y(1)] = 3$, $\mathbb{E}[Y(0)] = 1$, $\mathbb{E}[Y(1) \mid T{=}1] = 4$, $\mathbb{E}[Y(0) \mid T{=}1] = 2$, $\mathbb{E}[Y(1) \mid T{=}0] = 2$, $\mathbb{E}[Y(0) \mid T{=}0] = 0$.

(a) Compute the ATE and ATT. Are they equal?
(b) Compute the naïve estimator $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$. Decompose the gap between this and the ATE into a selection bias term and an ATT–ATE difference.
(c) Under what graphical condition on the DAG would the naïve estimator equal the ATE? State this condition in both potential outcome language (ignorability) and do-calculus language (back-door criterion).

3. Causal estimands from the structural equation. Consider the nonparametric SEM $Y = g(T, X, U_Y)$ with binary treatment $T \in \{0, 1\}$, observed covariate X, and unobserved error U_Y independent of (T, X) in the mutilated graph.

(a) Write τ_{ATE} and τ_{ATT} as expectations of $g(1, X, U_Y) - g(0, X, U_Y)$ over the appropriate distribution. Under what condition do they coincide?
(b) Specialize to the linear SEM $g(t, x, u) = \alpha + \beta t + \gamma x + u$. Show that the unit-level causal effect $Y_i(1) - Y_i(0)$ is constant across all units, and hence $\tau_{\text{ATE}} = \tau_{\text{ATT}} = \beta$.
(c) Now consider the heterogeneous SEM $g(t, x, u) = (\alpha + u)\, t + \gamma x$, where $U_Y \sim \mathcal{N}(0, \sigma^2)$, $\text{Cov}(U_Y, T) = \rho$, and $P(T{=}1) = p \in (0, 1)$. Compute τ_{ATE} and τ_{ATT}. Show that they differ when $\rho \neq 0$, and interpret this difference.
(d) In part (c), explain why OLS estimation of Y on T and X does not recover either τ_{ATE} or τ_{ATT} when $\rho \neq 0$.

4. SWIGs and ignorability. Consider the DAG: $X \to T$, $X \to Y$, $T \to Y$, with X fully observed.

(a) Construct the SWIG $\mathcal{G}(t)$ by splitting T into its random and fixed halves. Draw the result, labeling the random half, fixed half, and the potential outcome $Y(t)$.
(b) In $\mathcal{G}(t)$, identify all paths between the random half T and $Y(t)$. Apply the SWIG d-separation rules to determine which paths are blocked and which are open before conditioning.

(c) Use d-separation in $\mathcal{G}(t)$ to verify that $(Y(t) \perp\!\!\!\perp T \mid X)_{\mathcal{G}(t)}$ holds. Conclude that X satisfies the back-door criterion, confirming Theorem 4.1.

(d) Now add a hidden common cause $U \to T$, $U \to Y$. Draw the revised SWIG. Does the ignorability argument still hold? State the correct conclusion and identify what additional structure (if any) would be needed for identification.

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Appendix A

Graphical Intuition for Conditional Independence and d-Separation

This appendix provides a gentle introduction to the probabilistic and graphical ideas that underlie Chapters 2 and 3. Our goal is not to give a complete treatment of graphical models, but rather to develop enough intuition so that the formal machinery of d-separation, back-door adjustment, and intervention graphs does not appear abruptly.

A directed acyclic graph (DAG) is more than a picture. It is a compact language for expressing assumptions about how variables are related. Once those assumptions are represented graphically, the graph tells us which variables may be associated, which paths transmit dependence, and which variables should or should not be conditioned on.

The key pedagogical idea of this appendix is simple: before learning the full d-separation criterion, it is helpful to understand three elementary three-node patterns. Those local patterns explain most of what happens later in larger graphs.

A.1 Conditional Independence: The Probabilistic Language Behind Graphs

Before introducing graphs, we first recall the probabilistic notion that graphs are designed to encode.

Definition: Conditional Independence

Let X, Y, and Z be random variables. We say that X and Y are *conditionally independent given* Z, written $X \perp\!\!\!\perp Y \mid Z$, if

$$p(x, y \mid z) = p(x \mid z)\, p(y \mid z)$$

for all z with positive probability.

Equivalently, once Z is known, learning X gives no additional information about Y. In terms of densities, conditional independence admits the following equivalent characterizations:

$$X \perp\!\!\!\perp Y \mid Z \iff f(x, y, z)\, f(z) = f(x, z)\, f(y, z) \iff \exists\, a, b\colon f(x, y, z) = a(x, z)\, b(y, z).$$

The last form is especially useful: it says that the joint density factors into one piece depending on (x, z) and another depending on (y, z), with no cross-term in x and y.

Proposition: Fundamental Properties of Conditional Independence

(Dawid 1979, 1980) For random variables X, Y, Z, and W, the following hold:

- **(C1) Symmetry.** $X \perp\!\!\!\perp Y \mid Z \Rightarrow Y \perp\!\!\!\perp X \mid Z$.
- **(C2) Decomposition.** $X \perp\!\!\!\perp (Y, W) \mid Z \Rightarrow X \perp\!\!\!\perp Y \mid Z$.

- **(C3) Weak Union.** $X \perp\!\!\!\perp (Y, W) \mid Z \;\Rightarrow\; X \perp\!\!\!\perp Y \mid (Z, W)$.
- **(C4) Contraction.** $X \perp\!\!\!\perp Y \mid Z$ and $X \perp\!\!\!\perp W \mid (Y, Z) \;\Rightarrow\; X \perp\!\!\!\perp (Y, W) \mid Z$.
- **(C5) Intersection.** If $f(x, y, z, w) > 0$ for all (x, y, z, w), then $X \perp\!\!\!\perp Y \mid (Z, W)$ and $X \perp\!\!\!\perp Z \mid (Y, W) \;\Rightarrow\; X \perp\!\!\!\perp (Y, Z) \mid W$.

Properties (C1)–(C4) hold for any probability distribution and are known as the *semigraphoid axioms*. Property (C5) additionally requires the joint density to be strictly positive; together with (C1)–(C4) it forms the *graphoid axioms*. These properties are used implicitly throughout the course whenever conditional independence statements are combined or simplified.

Conditional independence is not the same as marginal independence. Two variables may be dependent marginally but independent after conditioning on a third variable. Conversely, two variables may be independent marginally but become dependent after conditioning. Both phenomena occur repeatedly in causal inference.

Example: Ice Cream, Drowning, and Season

Let X = ice cream sales, Y = drowning incidents, and Z = season. Marginally, X and Y are positively associated because both tend to be higher in summer. This does not mean that ice cream sales cause drowning. A more plausible explanation is that season is a common cause of both variables. Once season is fixed, the association largely disappears: $X \perp\!\!\!\perp Y \mid Z$.

This example illustrates a recurring theme in causal inference: an observed association may be induced by a third variable, and conditioning on that variable can remove the spurious dependence.

Remark: The Central Question

At this stage, the main question for the reader is: *Does conditioning on a variable remove association, preserve it, or create it?* Graphs provide a systematic answer to exactly this question.

A.2 Three Basic Motifs: Chain, Fork, and Collider

Every path in a DAG is built from local three-node configurations. There are three fundamental types: a *chain*, a *fork*, and a *collider*. Their behavior under conditioning is the foundation of d-separation.

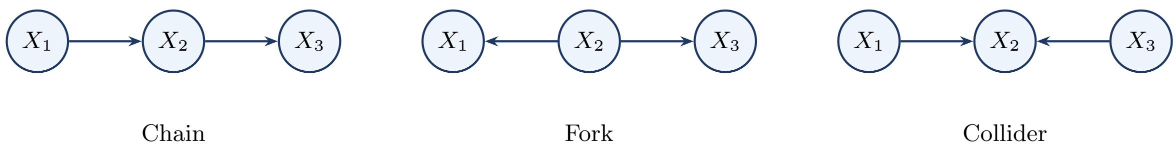

Figure A.1: The three fundamental three-node motifs. Chains and forks are blocked by conditioning on the middle node. Colliders are blocked by default but opened by conditioning on the middle node or one of its descendants.

A.2.1 Chain

Consider the pattern $X_1 \to X_2 \to X_3$, where the middle node X_2 lies on a directed pathway from X_1 to X_3.

Example: Exercise, Body Weight, and Blood Pressure

Let X_1 = exercise, X_2 = body weight, and X_3 = blood pressure. Exercise may affect blood pressure partly through its effect on body weight, so the path $X_1 \to X_2 \to X_3$ transmits association from X_1 to X_3. Marginally, exercise and blood pressure are associated; but if we condition on body weight, that particular pathway is blocked: $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

Once the middle variable is fixed, the chain no longer transmits additional information from X_1 to X_3.

A.2.2 Fork

Consider the pattern $X_1 \leftarrow X_2 \rightarrow X_3$, where the middle node X_2 is a common cause of X_1 and X_3.

> Example: Ice Cream, Season, and Drowning
>
> With X_1 = ice cream sales, X_2 = season, and X_3 = drowning incidents, season affects both variables, so X_1 and X_3 are associated even though neither causes the other. Conditioning on season blocks this path: $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

A fork is the simplest graphical form of confounding: the middle node creates association, and conditioning on it blocks the path.

A.2.3 Collider

Consider the pattern $X_1 \rightarrow X_2 \leftarrow X_3$, where the middle node X_2 is a common effect of X_1 and X_3.

> Example: Talent, Legacy Status, and College Admission
>
> Let X_1 = academic talent, X_3 = legacy status, and X_2 = admission to an elite university. Talent and legacy status may be unrelated in the general applicant pool. But among admitted students they can become statistically associated: low legacy status makes unusually high talent more likely among admitted applicants, and vice versa. Symbolically, $X_1 \perp\!\!\!\perp X_3$ but $X_1 \not\perp\!\!\!\perp X_3 \mid X_2$.

Unlike chains and forks, a collider *blocks* the path by default. Conditioning on the collider opens the path and may induce association that was not present marginally.

A.2.4 Summary of the Three Motifs

A chain ($X_1 \rightarrow X_2 \rightarrow X_3$) and a fork ($X_1 \leftarrow X_2 \rightarrow X_3$) are each open by default and blocked by conditioning on the middle node X_2. A collider ($X_1 \rightarrow X_2 \leftarrow X_3$) is blocked by default and opened by conditioning on the middle node or any of its descendants.

> Conditioning Is Not Always Beneficial
>
> Many adjustment mistakes arise from forgetting this asymmetry. Conditioning on a collider can create bias rather than remove it. The informal rule "control for more variables" is unsafe in causal inference; d-separation teaches a more precise lesson: condition on the *right* variables, not simply on many variables.

Chapter 2 develops these three motifs into the full d-separation criterion; see in particular Section 2.3 for the blocking rules and Section 2.4 for an extended treatment of collider bias.

A.3 d-Separation: When Does Conditioning Block a Path?

The three-node motifs explain what happens locally on a path. The next step is to extend this logic to a general DAG, where two variables may be connected by many paths.

> Definition: Path
>
> A *path* between two nodes is a sequence of distinct nodes such that each consecutive pair is connected by an edge, regardless of edge direction.

> Definition: Blocked Path and d-Separation
>
> A path is *blocked* by a conditioning set S if at least one of the following holds:
> 1. The path contains a chain or fork node that belongs to S, or
> 2. The path contains a collider such that neither the collider nor any of its descendants belongs

to S.
A path is *open* given S if it is not blocked. Two nodes X and Y are *d-separated* by a set S if every path between X and Y is blocked by S.

A.3.1 A Confounding Example

Consider the DAG with edges $X \to T$, $T \to Y$, and $X \to Y$. Here T is the treatment, Y is the outcome, and X is a pre-treatment covariate that affects both. There are two paths from T to Y: the directed causal path $T \to Y$, and the back-door path $T \leftarrow X \to Y$.

Without conditioning, the back-door path is open, so the observed association between T and Y mixes the causal effect with confounding. Conditioning on X blocks the fork $T \leftarrow X \to Y$, thereby isolating the causal path.

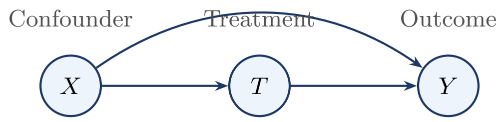

Figure A.2: A simple confounding graph. The path $T \leftarrow X \to Y$ is a back-door path from treatment to outcome. Conditioning on X blocks this noncausal path.

This confounding graph anticipates the back-door criterion of Chapter 3: the graphical condition that makes adjustment valid is precisely that all back-door paths are blocked.

A.3.2 A Collider Warning

Now consider the DAG with edges $T \to C$, $U \to C$, and $U \to Y$. Here C is a collider on the path $T \to C \leftarrow U \to Y$. Without conditioning on C, the path is blocked at the collider. Conditioning on C opens the path, creating a spurious association between T and Y through U.

Even more subtly, conditioning on a *descendant* of C can also open the path. If additionally $C \to D$, then conditioning on D may also induce association between T and Y through the collider at C.

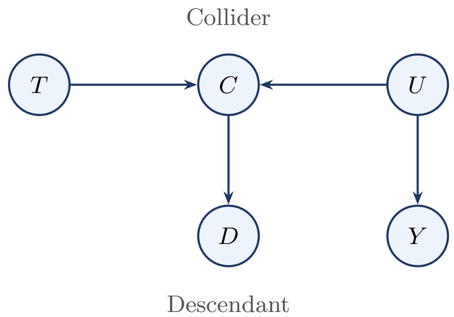

Figure A.3: Collider bias with a descendant. The path $T \to C \leftarrow U \to Y$ is blocked by default at the collider C. Conditioning on C, or on its descendant D, can open the path and induce a spurious association between T and Y.

A.3.3 A Practical Checklist

To decide whether X and Y are d-separated by S, proceed as follows. First, list all paths between X and Y. On each path, classify each interior node as part of a chain, fork, or collider. Then check whether each path is blocked by S. Finally, conclude that X and Y are d-separated if and only if every path is blocked.

For beginners, this path-by-path procedure is often more useful than starting with the most compact formal definition.

A.4 DAG Factorization and the Markov Property

Up to this point, we have used DAGs qualitatively, to decide which paths are open or blocked. We now connect the graph to probability algebra: the same parent structure that governs d-separation also determines how the joint distribution factorizes.

Definition: DAG Factorization

Let $\mathcal{G}$ be a DAG with nodes $V_1, \dots, V_p$. A joint distribution $p(v_1, \dots, v_p)$ is said to *factorize according to* $\mathcal{G}$ if

$$p(v_1, \dots, v_p) = \prod_{j=1}^{p} p\big(v_j \mid \mathrm{Pa}(V_j)\big)\,,$$

where $\mathrm{Pa}(V_j)$ denotes the set of parents of V_j in $\mathcal{G}$.

Example: A Simple Confounding Graph

For the DAG with edges $X \to T$, $T \to Y$, and $X \to Y$, the joint density factorizes as

$$p(x, t, y) = p(x)\, p(t \mid x)\, p(y \mid t, x).$$

Definition: Local Markov Property

A distribution P satisfies the *local Markov property* with respect to $\mathcal{G}$ if, for every node $V_i \in V$,

$$V_i \perp\!\!\!\perp (\mathrm{Nd}(V_i)\ \ \mathrm{Pa}(V_i)) \mid \mathrm{Pa}(V_i).$$

That is, once we condition on the direct causes of a node, that node is independent of all variables that are neither its descendants nor its parents.

Remark: Global Markov Property

The global Markov property is the statement that every d-separation in $\mathcal{G}$ implies a conditional independence in P. Thus, if two sets of variables are d-separated by a set S in the graph, then they are conditionally independent given S in the distribution. This is studied in detail in Chapter 2.

Remark: Equivalence for DAGs

For DAGs, the local and global Markov properties are equivalent under the usual graphical-model assumptions. The local property implies the global one via the Markov factorization, which is what is actually used in identification arguments.

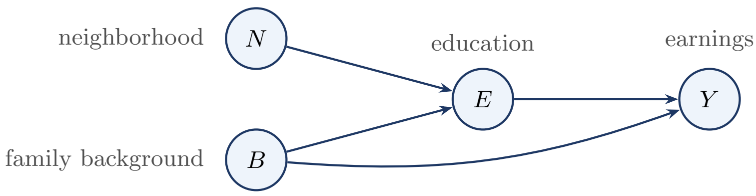

Figure A.4: Education–earnings DAG used to illustrate factorization and the local and global Markov properties.

Example: Education–Earnings Graph

Consider the DAG in Figure A.4, with edges $N \to E$, $B \to E$, $E \to Y$, and $B \to Y$. The corresponding factorization is

$$p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b).$$

The parent set of Y is $\mathrm{Pa}(Y) = \{E, B\}$, so the local Markov property gives $Y \perp\!\!\!\perp N \mid \{E, B\}$. The same conclusion follows from the global viewpoint: the only path from N to Y is $N \to E \to Y$, which is blocked by conditioning on E, and conditioning additionally on B opens no new path.

Remark: Edges as Scientific Claims

In a causal DAG, an arrow is typically drawn only when a direct dependence-generating relation is believed to be present. This explains why edges are not included gratuitously: every arrow represents a substantive scientific claim. Minimality and faithfulness are discussed in Chapter 2.

A.5 Optional: Moralization as an Alternative Criterion

Note to Reader

This section may be skipped on a first reading.

There is an alternative graph-theoretic way to check d-separation based on constructing an undirected graph called the *moral graph*. The method is elegant and useful in more advanced graphical-model theory, but it is not necessary for understanding the main causal ideas of Chapters 2 and 3. For most students, the path-by-path method using chains, forks, and colliders is more intuitive on first exposure.

The procedure is as follows. First, restrict attention to the ancestors of the variables of interest. Second, connect any two parents of a common child by an undirected edge. Third, drop all arrow directions. Finally, check ordinary graph separation in the resulting undirected graph. This procedure yields a criterion equivalent to d-separation.

Example: Moral Graph of a Collider

Suppose a DAG contains the pattern $A \to C \leftarrow B$. In the moral graph, the two parents A and B are connected by an undirected edge, producing the triangle $A - C - B$ together with $A - B$. This added edge reflects the special role of colliders in d-separation.

A.6 Summary

This appendix introduced the graphical ideas that support Chapters 2 and 3. Conditional independence is the probabilistic language that graphs are designed to encode. The three local motifs — chain, fork, and collider — determine how conditioning affects association along a path. D-separation extends these local rules to arbitrary graphs: two variables are d-separated by S if every path between them is blocked by S. The most important application in causal inference is to distinguish confounding paths from causal paths and to identify valid adjustment sets. Finally, the Markov property gives a probabilistic interpretation to the graph by linking graphical structure to a factorization of the joint distribution.